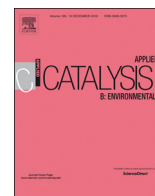




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journal homepage: www.elsevier.com/locate/apcatbEffects of initial crystal structure of Fe_2O_3 and Mn promoter on effective active phase for syngas to light olefinsYi Liu^a, Fangxu Lu^a, Yu Tang^b, Minyang Liu^a, Franklin Feng Tao^{b,*}, Yi Zhang^{a,*}^a State Key Laboratory of Organic-Inorganic Composites, Beijing University of Chemical Technology, 15 Beisanhuan East Road, Beijing 100029, PR China^b Department of Chemical and Petroleum Engineering, University of Kansas, Lawrence, KS 66045, USA

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ABSTRACT

Although it is generally acknowledged that iron carbides are the active phase in Fischer-Tropsch synthesis (FTS), adjusting the formation of active phase and catalytic performance via the initial crystal structure of iron oxides has never been studied. Herein, we take one-dimensional pure phase $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ nonporous nanorods to explore the effects of initial crystal phase of Fe_2O_3 on both formation of active phase and corresponding catalytic performance, in which the influence of diffusion limitation on selectivity can be ignored. *In situ* characterizations uncovered that the formation of iron carbides with high selectivity of light olefins strongly depends on the initial crystal phase of iron oxide and its induced metal-promoter interaction. Responding to the morphology effect and crystallographic phase effect, Mn located on surface of $\gamma\text{-Fe}_2\text{O}_3$ nanorods exhibit a low oxidation state due to a strong Fe-Mn interaction. This interaction was beneficial to the formation of C-poor iron carbide species at the metal-promoter interface during carburization. Hence, outstanding selectivity for light olefins (61.2%) was achieved on 0.5 Mn/ $\gamma\text{-Fe}_2\text{O}_3$ nanorods catalyst at a CO conversion of 55.1% under industrially relevant conditions (320 °C, 1 MPa, H_2/CO ratio of 1, and $\text{W/F} = 5 \text{ g}_{\text{cat}} \text{ h mol}^{-1}$). This finding enriches the fundamental understanding of active phase evolution and promoter effect in STO process and may guide the design of a catalyst with high selectivity.

1. Introduction

Fischer-Tropsch synthesis (FTS) has long been known as an alternative process to produce synthetic liquid fuels and valuable chemicals from coal, natural/shale gas and lignocelluloses biomass [1]. Recently, direct synthesis of light olefins ($\text{C}_2\text{-C}_4$) from syngas (STO) using Fe-based catalysts has become an increasingly important research topic in the conventional FTS process [2–4]. Generally, the most common initial phase of fresh Fe catalyst is present as iron oxides, which is hematite ($\alpha\text{-Fe}_2\text{O}_3$) and occasionally maghemite ($\gamma\text{-Fe}_2\text{O}_3$) [5–8]. And $\gamma\text{-Fe}_2\text{O}_3$ has the same XRD patterns with that of Fe_3O_4 [7], which might be ignored by previous researchers in FTS. The initial iron oxides are transformed to various types of iron carbides (such as $\chi\text{-Fe}_5\text{C}_2$, $\epsilon\text{-Fe}_2\text{C}$, and $\theta\text{-Fe}_3\text{C}$) [9–13] during reduction and carburization processes. The formation of different types of iron carbides was typically attributed to experimental kinetic factors (lattice deformation, carbon diffusion) and different carbon potential (μ_c) surroundings [10,14]. Although it is generally acknowledged that iron carbides are the active phase in FTS [15–17]. The correlation between initial phase of iron oxides and formation of iron carbide has never been studied and remained unknown. In fact,

maghemite ($\gamma\text{-Fe}_2\text{O}_3$) can be more easily reduced in contrast to $\alpha\text{-Fe}_2\text{O}_3$ phase [18,19]. As the formation of active phase of Fe-based catalyst of FT synthesis involves reduction of oxide to iron and carbonization of metal to carbide, we hypothesized that the initial surface chemistry and structure of the two different oxides ($\alpha\text{-Fe}_2\text{O}_3$ versus $\gamma\text{-Fe}_2\text{O}_3$) could significantly influence the formation of iron carbide phase. Thus, here we explored the potential correlation between properties of $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ and its phase transition to iron carbides in syngas and their corresponding catalytic performance. To clearly present the difference, we performed two sets of parallel studies by using pure $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ nonporous nanorods under same catalytic conditions, which can avoid the influence of diffusion limitation on the selectivity during the FTS reaction due to its nonporous structure and low surface area [11].

Manganese has been widely described as an effective promoter to increase olefins selectivity and suppress the formation of methane [20,21]; however, the promotion function is poorly understood since in most studies there was no control of other factors which may vary catalytic selectivity. In addition, iron oxide has been found to be a good candidate as cheap and efficient active metal and/or support, and it can form interaction with other active metal or promoter and enhance

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catalytic performance [22–24]. To clearly understand how the α -Fe₂O₃ and γ -Fe₂O₃ could be responsible for the promotion effect of Mn differently, we use pure phase α -Fe₂O₃ and γ -Fe₂O₃ nanorods which have the same morphology and very similar sizes as support to anchor Mn. Different from the conventional supported or bulk Fe nanoparticles catalysts with complicated composition and many mesopores or micropores, here pure phase α -Fe₂O₃ and γ -Fe₂O₃ nonporous nanorods were synthesized through an aqueous precipitation method to avoid the influence of other factors such as diffusion limitation [11,25], metal/promoter-inert support interaction [26], farraginous crystal facets [5], and ambiguous spatial collocation between Fe and Mn [27]. Based on various characterization results, we found that phase transitions of α -Fe₂O₃ and γ -Fe₂O₃ to iron carbides are distinctly different; the active phases from α -Fe₂O₃ and γ -Fe₂O₃ exhibit quite different selectivity of light olefins in FTS. On the other hand, due to the different properties of α -Fe₂O₃ and γ -Fe₂O₃, the more reduced MnO_{1-x} edge sites with strong Fe-Mn interaction was preferentially formed on the surface of γ -Fe₂O₃ nanorods, which was beneficial to the formation of C-poor iron carbides phase (θ -Fe₃C), resulting in forming more light olefins.

2. Experimental methods

2.1. Catalysts synthesis

The α -Fe₂O₃ and γ -Fe₂O₃ nanorods were prepared by an aqueous precipitation method as reported [28]. Briefly, a mixture containing FeCl₃·6H₂O (5.38 g), NaCl (11.60 g), and PEG (10 mL) was premixed and dissolved in water (190 mL) at 120 °C. A Na₂CO₃ aqueous solution (200 mL, 0.2 M) was then added under stirring at a rate of 5.5 mL min⁻¹. After that, the mixture was aged at 120 °C for 1 h. The precipitate was then washed with deionized water and ethanol, and finally dried at 50 °C for 6 h under vacuum to obtain the β -FeOOH precursor. The γ -Fe₂O₃ nanorods were prepared by heating the β -FeOOH precursor in PEG at 200 °C to reflux for 24 h under a stream of nitrogen. The resulting solid was washed with water and ethanol, followed by drying at 50 °C for 12 h under vacuum. The α -Fe₂O₃ nanorods were prepared by calcining the β -FeOOH precursor at 500 °C for 5 h. For preparation of Mn promoted Fe₂O₃ nanorods catalysts, in order to improving the dispersion of supported Mn [11], an ethylene glycol solution of Mn(NO₃)₂·4H₂O was impregnated onto the Fe₂O₃ nanorods, followed by drying at 200 °C under vacuum. All samples were pressed into pellets (10 MPa), crushed and sieved to retain 20–40 mesh particles for reaction tests.

2.2. Synthesis of iron carbide reference samples

The α -Fe₂O₃ precursors were firstly prepared by precipitation method. 15 g Fe(NO₃)₃·9H₂O (99.9%, Tianjing Chemical Co., China) were dissolved in 300 ml deionized water, and continuously stirred for 1 h at room temperature. Aqueous Na₂CO₃ (99.9%, Beijing Chemical Co., China) was used as a precipitating agent. Both solutions were simultaneously added to a precipitating batch containing 300 ml of distilled water at the start, and its rate was controlled in order to keep the pH of the slurry at 8.0 ± 0.2 during precipitation. The resulting precipitate was first filtered and then washed with deionized water. The precursor was dried in air at 120 °C for 12 h, and then calcined at 500 °C for 2 h. After that, χ -Fe₅C₂ reference materials was synthesized by exposing the precipitated α -Fe₂O₃ to a flow of 15 mL min⁻¹ CO (99.0%) and heating it up to 350 °C at 2 °C min⁻¹ [14].

θ -Fe₃C reference materials was synthesized by wet-chemical route [29]. The iron precursor solution was prepared by dissolving iron(II) acetate (1.5 g, Aldrich) in 20 mL deionized water. Gelatin (2 g, Type A from porcine skin, Sigma #G2500) was dissolved in deionized water (100 mL) at 90 °C to form a clear biopolymer solution. To synthesize iron carbide, the iron acetate solution (10 mL) was added dropwise to the biopolymer solution (6 mL) with fast stirring. The products were

dried at 80 °C in air for 5 h, and then calcined at 650 °C under a flow of N₂ for 2 h.

2.3. Catalysts characterization

For characterization of carburized catalysts, the catalysts were pre-carburized at 340 °C for 10 h using syngas (H₂/CO = 1) and then passivated by 1% O₂ in N₂ at room temperature to form a metal oxide layer on the surface of metal particle, in order to preventing oxidation of iron metal when the catalysts were exposed to air. Before the *in situ* XPS/XAFS characterization, the passivated samples were carburized again *in situ* at 340 °C for 1–2 h using syngas (H₂/CO = 1).

The size and morphology of the products were characterized by scanning electron microscope (SEM, JEOL, JSM-7600 F) and high-resolution transmission electron microscope (HRTEM, JEOL, JEM-2100). The specimen was prepared by ultrasonically suspending the sample in ethanol. A drop of the suspension was deposited on a coverslip or carbon-coated copper grids and dried in air. XRD measurements were determined on an X-ray diffractometer (SHIMADZU XRD-6000) with Cu K α radiation (λ = 0.154 nm), operated at 40 kV and 30 mA. Nitrogen adsorption isotherms measurement was carried out at -196 °C with a Micrometrics ASAP 2020 equipment. The sample was degassed at 180 °C for 6 h prior to the measurement. Raman spectroscopy (Renishaw UV-vis Raman 1000) with laser excitation at a visible 514 nm (Ar⁺ laser) was conducted at room temperature to evaluate the structure of samples.

In-situ XPS experiments were performed at the photoemission end-station at beamline BL10B in the National Synchrotron Radiation Laboratory (NSRL) in Hefei, China. The beamline is connected to a bending magnet and covers photon energies from 60 to 1000 eV with a resolving power (E/ Δ E) better than 1000. The end-station is composed of four chambers, *i.e.*, analysis chamber, preparation chamber, quick sample load-lock chamber and high pressure reactor. The analysis chamber, with a base pressure of < 2 × 10⁻¹⁰ torr, is connected to the beamline and equipped with a VG Scienta R3000 electron energy analyzer and a twin anode X-ray source. The high pressure reactor houses a reaction cell where the samples can be treated with different gases up to 20 bar and simultaneously heated up to 650 °C. After the sample treatment, the reactor can be pumped down to high vacuum (< 10⁻⁸ torr) for sample transfer. In the current work, the pre-carburized samples were pressed into thin pellets and then placed into the reaction cell and carburized in a flowing syngas (H₂/CO = 1) under atmospheric pressure at 340 °C for 2 h. Then the samples were cooled down to room temperature and transferred to the analysis chamber under vacuum conditions for measurements.

H₂ temperature programmed reduction (H₂-TPR) experiments were carried out in a quartz tube reactor using 0.05 g as-prepared catalysts. The reducing gas, a mixture of 10% H₂ diluted by Ar, was fed via a mass flow controller at 30 mL min⁻¹ and the temperature was increased from 30 °C until 800 °C at a rate of 8 °C min⁻¹. The effluent of reactor passed through a 5 Å molecular sieve trap to remove produced water, before reaching TCD. CO-TPD experiments were performed on Micrometrics 2020 chemisorption instrument equipped with a thermal conductivity detector (TCD), using Helium as the carrier gas. The sample loading was 0.1 g. The pretreatment conditions were the same as the reduction conditions before FTS reaction. After pretreatment, the sample was cooled to 50 °C and exposed to 5% CO/95% He (30 mL min⁻¹) for 30 min, purged in He for 1 h at 50 °C and then heated linearly at 10 °C min⁻¹ to 800 °C.

In situ X-ray absorption spectra at Fe K-edge and Mn K-edge for the catalysts and the reference compounds were measured on the beamline 1W1B of Beijing Synchrotron Radiation Facility (BSRF). The storage ring of BSRF was operated at 2.5 GeV with a maximum current of 120 mA. The X-ray absorption spectra at K-edge of the metals were collected in transmission mode. Fe foil (99.99%), α -Fe₂O₃ (99.9%, Aldrich-sigma) and γ -Fe₂O₃ (99.9%, Aldrich-sigma) were used as

references. The XAS data was processed and analyzed by Demeter software package including Athena and Artemis [30]. The k^2 -weighted r -space spectrum was obtained from Fourier transformation and then fitted. The fitting range of k is 3–10 Å⁻¹, while the range of r -space is 1–4 Å. Coordination number, distance and Debye-Waller factor (ss) were fitted as variables. The iron phase compositions were fitted from XANES by using principle combination analysis (PCA) and the linear combination fitting (LCF) method.

Transmission ⁵⁷Fe Mössbauer spectra were collected at room temperature on an MR-351 constant-acceleration Mössbauer spectrometer (FAST, Germany), using a 25 mCi ⁵⁷Co in Pd matrix. The spectrometer was operated in the symmetric constant acceleration mode. Data analysis was performed using a nonlinear least-squares fitting routine that models the spectra as a combination of singlets, quadruple doublets, and magnetic sextuplets based on a Lorentzian line shape profile.

2.4. Fischer-Tropsch synthesis reaction

The FTS reaction was carried out in a continuous-flow-type fixed-bed reactor (stainless steel 316, 8 mm i.d.) under the temperature of 320 °C and a pressure of 1.0 MPa. The feed gas consisted of Ar (4.9%), CO (47.5%) and H₂ (47.6%), where Ar was used as inner reference gas. The 0.5 g catalyst mixed with 1.0 g quartz sand was placed in the reactor tube. The temperature was then increased from room temperature to 340 °C at a rate of 2 °C·min⁻¹. The catalyst was carburized *in situ* at 340 °C for 10 h in the flow of feed gas (W/F(CO + H₂ + Ar) = 2.5 g_{cat}·h·mol⁻¹) at 0.1 MPa, followed by cooling down to 80 °C in N₂. After that, the reaction temperature was increased to 320 °C at a rate of 10 °C·min⁻¹, with a flow of pressurized syngas (W/F(CO + H₂ + Ar) = 5 g_{cat}·h·mol⁻¹). The reaction was continuously carried out for 20–50 h.

The effluent gas from the reactor was analyzed by on-line gas chromatography (GC-2014C, Shimadzu). A thermal conductivity detector (TCD) was used to analyze gaseous products (CO, CO₂ and CH₄). Light hydrocarbons (C₁–C₅) were on-line analyzed by another GC with flame ionization detector (FID) using a Porapak-Q column. The analyses of hydrocarbons dissolved in the solvent (10 ml *n*-decane) and cooled in the trap were carried out with GC-FID using silicone SE-30 column. The mass balance of various reactions were calculated, which were almost 95% based on carbon mole for all reactions.

3. Results and discussion

3.1. Relationship between initial crystal phases and catalytic performances

Fig. 1a is a typical X-ray powder diffraction (XRD) patterns of the as-prepared α-Fe₂O₃ and γ-Fe₂O₃ nanorods, suggesting that the as-synthesized sample is well crystallized α-Fe₂O₃ (JCPDS 33-0664) and γ-Fe₂O₃ (JCPDS 39-1346) phase, respectively. As γ-Fe₂O₃ has the same XRD patterns with that of Fe₃O₄, the formation of γ-Fe₂O₃ phase was

further confirmed by the appearance of broad Raman characteristic features around 353, 491, 665, and 720 cm⁻¹ (Fig. 1b) [18,31] and XANES spectra (Fig. S1). Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) observations (Fig. 2) revealed that the as-prepared α-Fe₂O₃ and γ-Fe₂O₃ samples have the same morphology, which are nanorods with an average diameter of 40 nm and length of 500 nm. However, they have distinctly different crystal phases. The measured interplanar distances of 0.250 nm and 0.295 nm in HRTEM (Fig. 2c and f) corresponds to the (110) plane of α-Fe₂O₃ and (220) plane of γ-Fe₂O₃, respectively. Moreover, both α-Fe₂O₃ and γ-Fe₂O₃ display type-II N₂ sorption isotherm (Fig. S2) typically associated with nonporosity, yielding the low surface area of 11.7 m² g⁻¹ and 12.5 m² g⁻¹, respectively, those could guarantee to avoid the influence of diffusion limitation on selectivity during FTS reaction. Thus, the successful preparation of α-Fe₂O₃ and γ-Fe₂O₃ with highly similar size and surface area allow exploring crystal phase-dependent formation of iron carbide and correlating this phase formation and catalytic selectivity.

The as-prepared pure α-Fe₂O₃ and γ-Fe₂O₃ nanorods were carburized at 340 °C for 10 h and then tested in the FTS at 320 °C, 1 MPa, H₂/CO ratio of 1, and W/F = 5 g_{cat}·h·mol⁻¹. As shown in Fig. 3a, b, The CO conversion and light olefins selectivity of both catalysts increased during the first 5 h of reaction, suggesting an incubation of FTS. It is reasonable since this is the period for the formation of stable active phase. The α-Fe₂O₃ nanorods catalyst shows a lower maximum CO conversion (56.1%) than γ-Fe₂O₃ nanorods catalyst (82.5%), likely due to the lower reducibility of α-Fe₂O₃, as confirmed by H₂-TPR result (Fig. S3). Mössbauer spectra results of the reduced samples (Table 2) also show that the concentration of Fe_xC in γ-Fe₂O₃ catalyst is higher than that in α-Fe₂O₃ catalyst, which contributed to the higher catalytic activity. After FTS of 20 h on stream, the catalytic performances reached a steady state (Fig. 3a, b) and the CO conversion of α-Fe₂O₃ and γ-Fe₂O₃ catalysts were change to 52.7% and 57.1%, respectively. Meanwhile, it was found that α-Fe₂O₃ catalyst exhibited much higher light olefins selectivity (44.5%) and olefin/paraffin ratio in C₂–C₄ (4.01) compared with γ-Fe₂O₃ nanorods catalyst. The γ-Fe₂O₃ catalyst exhibits a light olefins selectivity of only 25.4% with olefin/paraffin ratio of 1.09 (Fig. 3c and Table 1). Thus, the observed distinct difference in selectivity must result from the potentially different phase transitions, which resulted in the formation of different active phases.

To reveal the correlation between initial phase and the formation of active phase, multiple characterization techniques were used to track surface and bulk structures of Fe catalyst. Based on the XRD studies of carburized (Figs. S4–S5) and spent catalysts (Fig. S6), Hägg carbide (χ-Fe₅C₂, JCPDS 36-1248) was the only carbide species detected from carburized/spent γ-Fe₂O₃ catalyst. In contrast, for α-Fe₂O₃ catalyst, a broad diffraction peak centered at ~45° indexed to cementite (θ-Fe₃C, JCPDS 76-1877) was observed after carburization and 20 h reaction. The identification of iron carbide phases with XRD was further

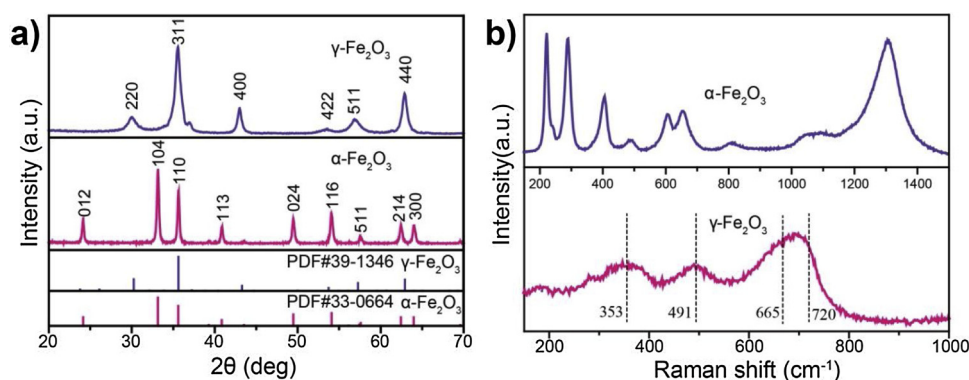


Fig. 1. a) XRD patterns for the as-prepared α-Fe₂O₃ and γ-Fe₂O₃ nanorods. b) Raman spectra of the as-prepared α-Fe₂O₃ and γ-Fe₂O₃ nanorods.

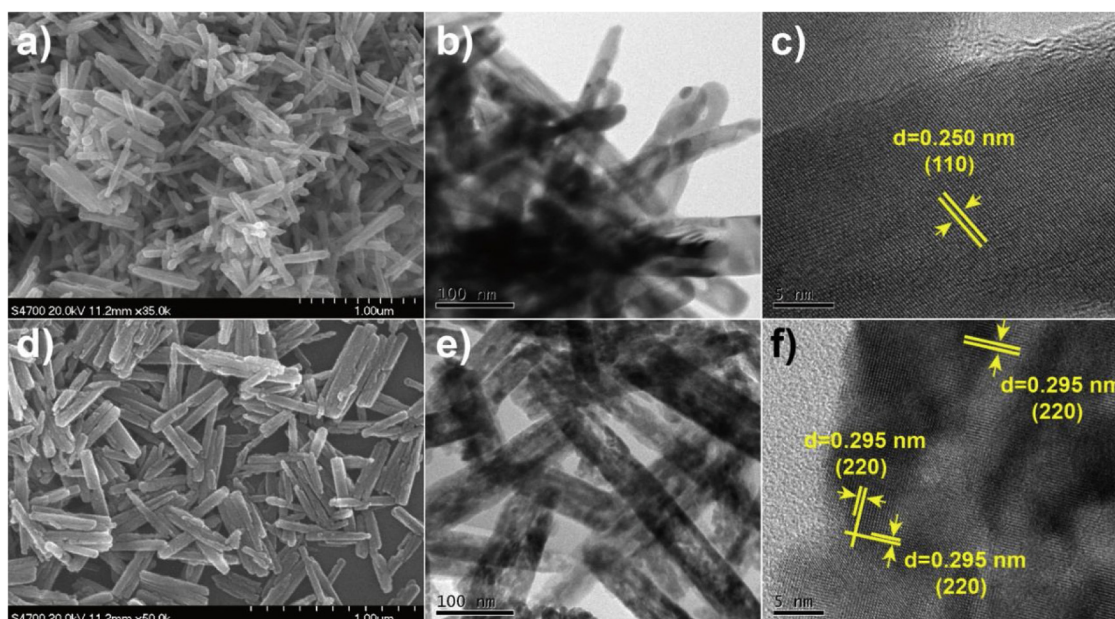


Fig. 2. a) SEM image of the α - Fe_2O_3 nanorods. b–c) HRTEM images of the α - Fe_2O_3 nanorods. d) SEM image of the γ - Fe_2O_3 nanorods. e–f) HRTEM images of the γ - Fe_2O_3 nanorods.

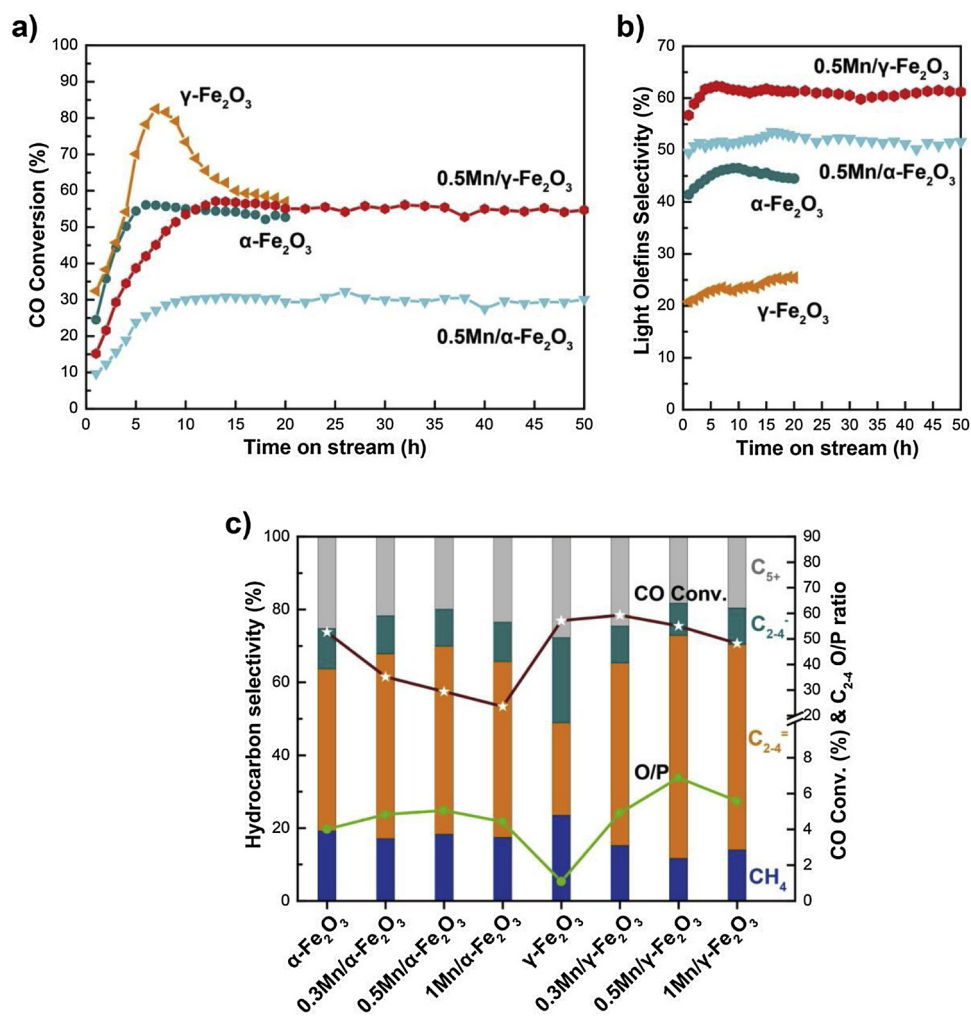


Fig. 3. a) CO conversion and stability of various iron-based catalysts; b) Light olefins selectivity as a function of time on-stream for various iron-based catalysts; c) Catalytic performance of all Fe_2O_3 nanorods catalysts with and without Mn promoter. Reaction conditions: $\text{H}_2/\text{CO} = 1$, $T = 320^\circ\text{C}$, $P = 1.0\text{ MPa}$ and $W/F(\text{CO} + \text{H}_2 + \text{Ar}) = 5\text{ g}_{\text{cat}}\cdot\text{h mol}^{-1}$.

Table 1
Catalytic activity and product selectivity of various iron-based catalysts.^a

Entry	Catalysts	CO conv. (%) ^b	CO conv. (%) ^c	CO ₂ (%CO conv.) ^c	Hydrocarbon selectivity (c-mol %, CO ₂ -free) ^c				O/P ^d
					CH ₄	C ⁼ ₂₋₄	C ₂₋₄	C ₅₊	
1	γ-Fe ₂ O ₃	82.5	57.1	34.1	23.5	25.4	23.4	27.7	1.09
2	0.3 Mn/γ-Fe ₂ O ₃	64.0	59.3	30.4	15.2	50.1	10.2	24.5	4.91
3	0.5 Mn/γ-Fe ₂ O ₃	57.1	55.1	31.7	11.7	61.2	8.9	18.2	6.88
4	1Mn/γ-Fe ₂ O ₃	52.4	48.2	33.9	14	56.4	10.1	19.5	5.58
5	α-Fe ₂ O ₃	56.1	52.7	28.0	19.2	44.5	11.1	25.2	4.01
6	0.3 Mn/α-Fe ₂ O ₃	38.7	35.2	22.5	17.1	50.7	10.5	21.7	4.83
7	0.5 Mn/α-Fe ₂ O ₃	30.8	29.4	19.1	18.3	51.6	10.2	19.9	5.06
8	1Mn/α-Fe ₂ O ₃	24.9	23.6	20.2	17.5	48.2	10.9	23.4	4.42

^a Reaction condition: 320 °C, 1.0 MPa, H₂/CO = 1, W/F(CO + H₂ + Ar) = 5 g_{cat} h mol⁻¹, time on stream 20 or 50 h.

^b The highest conversion with time on stream.

^c The data were taken after 20 h of time on stream.

^d The molar ratio of olefin to paraffin in the fraction C₂-C₄.

Table 2
Mössbauer parameters of the carburized catalysts after 10 h of carburization at T = 340 °C, P = 0.1 MPa and H₂/CO ratio of 1.

Catalysts	Phases	Mössbauer parameters ^a				
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	Area ^b (%)	
α -Fe ₂ O ₃	Fe ₅ C ₂	0.40	0.80	225	5.0	
		0.01	−0.61	185	5.0	
		0.06	0.13	118	2.5	
	Fe ₃ C	0.22	−0.17	208	8.4	
		0.31	0.53	204	4.2	
	Fe ₃ O ₄	0.53	−0.005	486	34.9	
		0.59	0.26	446	33.7	
	Fe ³⁺ (spm)	0.40	−1.45		0.9	
	Fe ²⁺ (spm)	0.70	−1.43		5.4	
	γ -Fe ₂ O ₃	Fe ₅ C ₂	0.40	0.80	212	12.0
0.10			−0.55	192	12.0	
0.04			0.80	117	5.8	
Fe ₃ O ₄		0.37	−0.04	508	24.2	
		0.37	−0.02	459	30.6	
Fe ³⁺ (spm)		0	−0.98		4.8	
Fe ²⁺ (spm)		0.70	−0.95		10.6	
0.5 Mn/ α -Fe ₂ O ₃		Fe ₅ C ₂	0.03	0.56	224	6.3
			0.14	0.27	192	6.3
			0.26	−0.15	113	3.2
	Fe ₃ C	0	−0.80	210	2.4	
		0.19	−0.80	204	1.2	
	Fe ₃ O ₄	0.30	−0.01	493	32.8	
		0.60	0.05	454	31.6	
	Fe ³⁺ (spm)	0.16	−0.85		8.0	
	Fe ²⁺ (spm)	1.16	−0.51		8.2	
	0.5 Mn/ γ -Fe ₂ O ₃	Fe ₅ C ₂	0.12	−0.80	198	4.3
0.23			−0.50	180	4.3	
0.21			−0.05	110	2.1	
Fe ₃ C		0.18	−0.04	208	14.0	
		0.34	0.80	207	7.0	
Fe ₃ O ₄		0.49	0.64	475	20.5	
		0.41	−0.34	484	36.4	
Fe ³⁺ (spm)		0.02	0.50		6.8	
Fe ²⁺ (spm)		1.20	−0.75		4.6	

^a Measured at room temperature.

^b Max error = ± 1%.

supported by HRTEM images and Mössbauer spectra. HRTEM images (Fig. 4c-f) showed that the Fe₂O₃ species were converted to Fe₃O₄ and Fe₅C₂ phase as a result of carburized treatment for both samples. In the case of the α-Fe₂O₃ sample, Fe₃C nanoparticles with exposed facets of (121) were observed as shown in Fig. 4c. In addition, it was found that the nanorod structure of α-Fe₂O₃ and γ-Fe₂O₃ both maintained very well after carburization (Fig. 4a, b) and 20 h reaction (Fig. 4g, h).

Mössbauer spectroscopy is most powerful tool for identifying and quantifying iron phases in iron-based catalysts [12,13]. In the

Mössbauer spectrum of carburized α-Fe₂O₃ catalyst (Fig. S7 and Table 2), three sextets (225, 185, and 118 kOe) for χ-Fe₅C₂ and two sextets (208 and 204 kOe) for θ-Fe₃C were observed [32], showing coexistence of both χ-Fe₅C₂ and θ-Fe₃C on FTS catalyst formed from α-Fe₂O₃; the total concentration of Fe_xC (χ-Fe₅C₂ + θ-Fe₃C) formed from α-Fe₂O₃ is 25.1%. In contrast, only χ-Fe₅C₂ was formed from γ-Fe₂O₃ and its fraction is 29.8%. We further explored Mössbauer spectra of α-Fe₂O₃ and γ-Fe₂O₃ catalysts after 20 h on stream in FTS (Fig. 5 and Table 3). After reaction, the content of iron oxide in α-Fe₂O₃ catalyst decreases from 74.9% in carburized catalyst to 66.9%, indicating that further carburization takes place during the reaction. The content of χ-Fe₅C₂ increases to 22.3%, while the content of Fe₃C slightly decreases to 10.8%. The further carburization during the reaction is probably due to the higher gas pressure (1 MPa) and longer reaction time compared with the carburization step. In contrast, the content of χ-Fe₅C₂ in the spent γ-Fe₂O₃ catalyst also increased to 34.0%, which is higher than the value of 29.8% in the carburized γ-Fe₂O₃ catalyst. However, the θ-Fe₃C phase is still absent in spent γ-Fe₂O₃ catalyst. In addition, for both carburized and spent α-Fe₂O₃ or γ-Fe₂O₃ samples, the Mössbauer spectra include two doublets, which is assigned to the superparamagnetic (spm) Fe³⁺/Fe²⁺ ions with the crystallite diameters smaller than 13.5 nm [2,6]. This is due to the limitation of room temperature Mössbauer spectra. At lower measurement temperatures (< 25 K), the coupling forces prevail over the thermal energy, causing the magnetic moments of neighbouring atoms to align and the Mössbauer spectra becomes magnetically split (hyperfine sextuplets are observed) [2].

Fig. S8 shows the adsorption behavior of CO on carburized α-Fe₂O₃ and γ-Fe₂O₃ catalysts determined by CO-TPD. The Mössbauer spectra results (Table 2) mentioned above indicate that carburization in syngas leads mainly to the formation of Fe₃O₄ and Fe_xC species. Therefore, the peaks of CO desorption at 380 and 440 °C can be attributed to the desorption of surface and/or near-surface C and O species on Fe₃O₄ and Fe_xC, respectively [6,33]. Furthermore, there was another CO desorption peak on Fe_xC at 490 °C for α-Fe₂O₃ sample, demonstrating multiple CO adsorption sites. Such a result possibly originated from the Fe₅C₂ and Fe₃C mixing phase in carburized α-Fe₂O₃ catalyst [34], as confirmed by Mössbauer spectra, XRD, and HRTEM results. Meanwhile, the amount of adsorbed CO on γ-Fe₂O₃ sample was much larger than that on α-Fe₂O₃, showing the higher CO adsorption capacity on γ-Fe₂O₃ catalyst.

The different phase transitions from iron oxides suggest that C-poor θ-Fe₃C can preferentially form from α-Fe₂O₃ phase. We speculate that the formation of C-rich iron carbides (χ-Fe₅C₂) on γ-Fe₂O₃ instead of C-poor θ-Fe₃C is due to the higher carburization ability of γ-Fe₂O₃. The ease of being carburization on γ-Fe₂O₃ is due to a ready cleavage of Fe-O bond of an inverse spinel structure [35]. In contrast to γ-Fe₂O₃, FTS

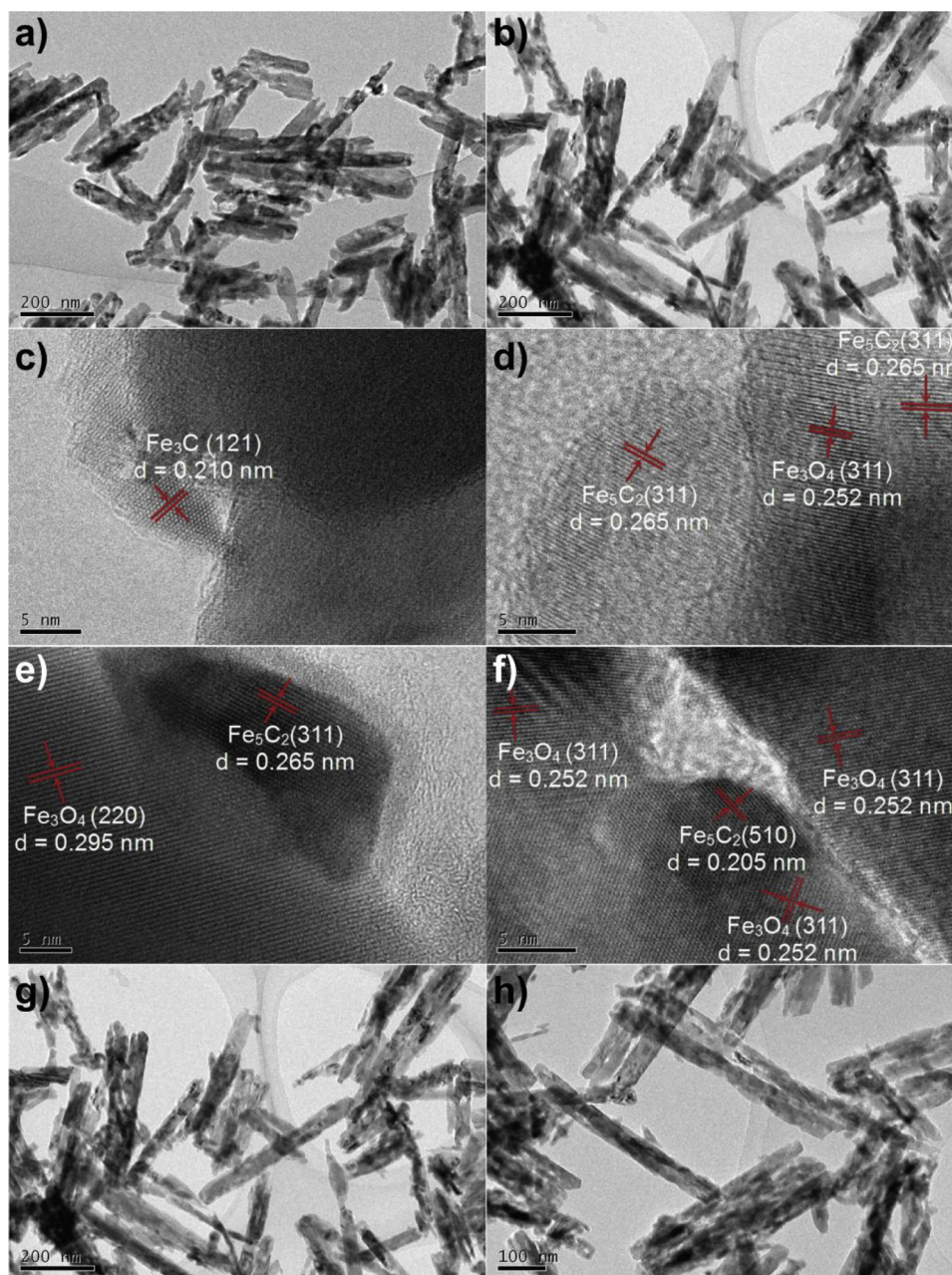


Fig. 4. HRTEM of α -Fe₂O₃ (a, c, e, g) and γ -Fe₂O₃ (b, d, f, h) nanorods after carburized (a–f) and after FTS reaction (g, h). Carburized conditions: 340 °C, 0.1 MPa, H₂/CO = 1, W/F(CO + H₂ + Ar) = 2.5 g_{cat}·h·mol⁻¹. Reaction conditions: 320 °C, 1.0 MPa, H₂/CO = 1, W/F(CO + H₂ + Ar) = 5 g_{cat}·h·mol⁻¹.

over α -Fe₂O₃ catalyst exhibited higher light olefins selectivity with much higher O/P ratio in products (Fig. 3c). This is consistent with earlier studies which revealing that θ -Fe₃C is responsible for the formation of light olefins [11,36], as it has relatively weak hydrogenation activity for unsaturated hydrocarbon [37].

Chemical and coordination environments of Fe in precursors and carburized/spent state were studied with *in situ* manner as reported [14,38]. The nonphase corrected, k^2 or k^3 weighted Fourier transformed Fe K-edge EXAFS spectra for the fresh, carburized, and spent catalysts are shown in Figs. 6 and 7a. As shown in Fig. 6, the fresh α -Fe₂O₃ and γ -Fe₂O₃ catalyst shows three distinct scattering peaks (1.45, 2.55, and 3.16 Å), characteristics of α -Fe₂O₃ or γ -Fe₂O₃ [12]. After carburized/reaction, a new peak at ~ 2.00 Å (value before phase correction, the same in the following) appeared, which is contributed from the first shell of Fe-(C)-Fe structure of iron carbides [8,12]. In addition, the different shape and broadening of the shell at 1.45 Å for carburized

catalysts indicate that iron oxide and iron carbide species coexist, as both Fe-C and Fe-O bonds affect this peak. As reported in literature [9–13], θ -Fe₃C and χ -Fe₅C₂ can be distinguished with the fact that the first distance of Fe and C in Fe-(C)-Fe structure of χ -Fe₅C₂ is longer than that of θ -Fe₃C, since the Fe lattice of χ -Fe₅C₂ has to expand to accommodate more C atoms than carbon-poor θ -Fe₃C. Reference samples θ -Fe₃C and χ -Fe₅C₂ was synthesized by wet-chemical route and *in situ* carbonization of hematite, respectively (Figs. S9–10); the r-space spectrum of θ -Fe₃C and χ -Fe₅C₂ were presented in Fig. S11. The peak center of the Fe-Fe shell of χ -Fe₅C₂ is 2.16 Å, while it is 1.97 Å for θ -Fe₃C. For the carburized/spent α -Fe₂O₃ catalyst, it is proposed that the shorter Fe-Fe shell of iron carbides (Figs. 6 and 7a and Table 4) is contributed by the coexistence of θ -Fe₃C and χ -Fe₅C₂ carbide phase, as also confirmed by XRD, HRTEM, and Mössbauer spectra.

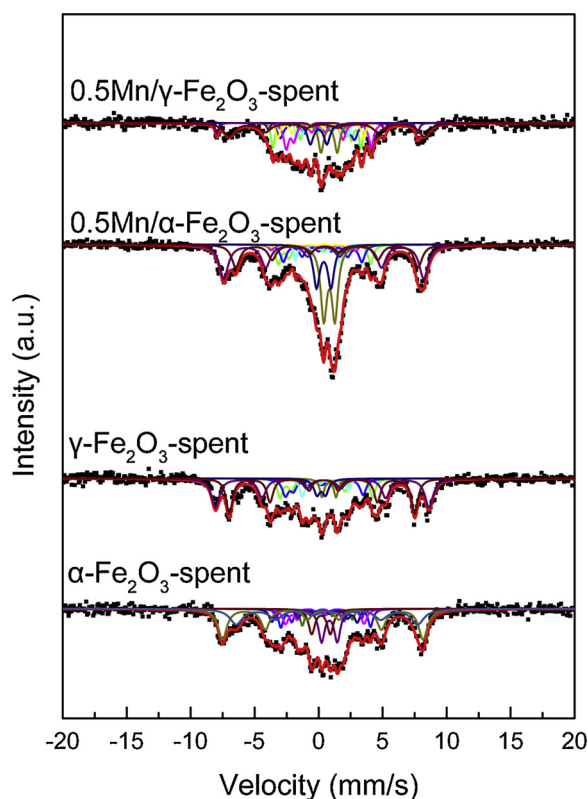


Fig. 5. Mössbauer spectra of α -Fe₂O₃ and γ -Fe₂O₃ based nanorods catalysts after 20 h reaction.

3.2. Correlation between original iron oxide structure and Mn promotion effect

Mn is one of the main elements, which promote catalytic selectivity of olefins in FTS. Hence, we further explored Mn-modulated iron oxide nanorods catalysts by impregnation method (Fig. S12) to improve FTS catalytic performance. Our catalysis studies of Mn/ γ -Fe₂O₃ and Mn/ α -Fe₂O₃ showed that γ -Fe₂O₃ and α -Fe₂O₃ responded to the promotion of Mn to light olefins selectivity distinctly differently (Fig. 3 and Table 1). In the case of 0.5 Mn/ γ -Fe₂O₃, addition of Mn to γ -Fe₂O₃ results the increase of light olefins selectivity from 25.4% to 61.2%, while the conversion of CO slightly decreased. Olefin/paraffin ratio for C₂-C₄ (O/P) is another important parameter for evaluate the extent of olefin secondary reaction to paraffins. In terms of 0.5 Mn/ γ -Fe₂O₃, it also exhibits much higher O/P ratio (O/P = 6.88) than nonpromoted γ -Fe₂O₃ catalyst (O/P = 1.09). Different from the significant promotion of Mn on γ -Fe₂O₃ catalysts, the improvement of light olefins selectivity is not significant for Mn-modulated α -Fe₂O₃ catalysts. Upon loading 0.5 wt. % Mn to α -Fe₂O₃, the selectivity for light olefins only increased from 44.5% to 51.6%, while CO conversion decreased from 56.1% to 30.8%. The Mössbauer spectra results (Tables 2 and 3) show that the Mn-promoted α -Fe₂O₃ samples exhibit much lower total carbides contents after reduction/reaction compared to non-promoted catalyst, and lower degree of carburization for Mn-promoted α -Fe₂O₃ catalysts could directly result in relatively lower catalytic activities. Table S1 shows LCF results of spent catalysts from XANES spectra. The contents of iron carbides fitted from XANES are consistent with the Mössbauer spectra results.

The CO₂ selectivity was defined as the percentage of CO that is converted to CO₂ in the total consumed CO. As seen in Table 1, the CO₂ selectivity of all catalysts was proportional to the FTS reaction activity. All γ -Fe₂O₃ based catalysts exhibited higher CO₂ selectivity than α -Fe₂O₃ based catalysts. This can be ascribed to the increased concentration of H₂O at higher CO conversion, which was believed to bring

Table 3

Mössbauer parameters of the spent catalysts after 20 h on stream at T = 320 °C, P = 1 MPa and H₂/CO ratio of 1.

Catalysts	Phases	Mössbauer parameters ^a			
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	Area ^b (%)
α -Fe ₂ O ₃	Fe ₃ C ₂	0.24	0.63	218	8.9
		0.30	0.36	191	8.9
		0.29	0.73	123	4.5
	Fe ₃ C	0	-0.64	207	7.2
		0.17	-0.29	208	3.6
γ -Fe ₂ O ₃	Fe ₃ O ₄	0.35	-0.04	487	27.8
		0.70	-0.11	441	20.5
	Fe ³⁺ (spm)	0.19	-1.45		8.1
	Fe ²⁺ (spm)	0.84	1.23		10.5
	Fe ₃ C ₂	0.28	0.65	225	14.1
		0.11	0.65	189	14.1
	Fe ₃ O ₄	0.19	0.62	114	5.8
		0.34	-0.12	519	27.5
	Fe ³⁺ (spm)	0.33	-0.13	450	26.5
		0.20	-0.67		6.4
0.5 Mn/ α -Fe ₂ O ₃	Fe ₃ C ₂	0.81	-1.10		5.6
		0.12	0.69	224	8.9
	Fe ₃ C	0.40	-0.18	191	8.9
		0.26	-0.54	111	4.4
	Fe ₃ O ₄	0	-0.80	206	2.7
		0	-0.80	204	1.3
	Fe ³⁺ (spm)	0.40	0.01	485	26.3
		0.57	0.15	443	16.6
	Fe ²⁺ (spm)	0.40	1.13		11.4
		0.84	0.87		19.5
0.5 Mn/ γ -Fe ₂ O ₃	Fe ₃ C ₂	0.22	-0.70	214	12.6
		0.21	-0.61	180	12.6
	Fe ₃ C	0.09	0.37	124	6.3
		0.39	0.80	206	16.5
	Fe ₃ O ₄	0	0.33	208	8.3
		0.25	-0.72	487	4.6
	Fe ³⁺ (spm)	0.43	0.37	474	20.2
		0	-1.29		8.7
	Fe ²⁺ (spm)	0.82	1.27		10.2

^a Measured at room temperature.

^b Max error = $\pm$ 1%.

about the intensification of WGS (water-gas shift) reaction [12]. It has been reported that the WGS reaction activity could affect the hydrocarbon selectivity because the increased H₂/CO ratio, leading to an increase in CH₄ selectivity and a decrease in longer hydrocarbon selectivity [39]. However, our present results suggested that there is no direct correlation between the CO₂ selectivity and the hydrocarbon selectivity, especially light olefins selectivity. This conclusion was also reported by other researchers [12,33].

In addition, further increase in the Mn concentration in the Mn/ γ -Fe₂O₃ or Mn/ α -Fe₂O₃ catalyst both led to an increase in methane selectivity and decrease in light olefins selectivity (Fig. 3c). We proposed that the surface Fe_xC sites of Mn/ γ -Fe₂O₃ and Mn/ α -Fe₂O₃ can be partitioned into two types of ensembles: region of unpromoted sites that are not close to MnO_x and region of promoted sites that are adjacent to MnO_x. The volcano-like evolution of selectivity as a function of Mn loading suggests that the regions of promoted sites are the interfacial area of Fe_xC_y patch with MnO_x clusters. This assignment is similar to the attribution of active site of WGS catalysis on Cu/CeO₂ to the interfacial area of Cu nanoparticles and CeO₂ [40–42]. At a relatively low loading (< 0.5 wt. %), the product selectivity increases along the fraction of the areas of ideal sites (Fe_xC_y patch with MnO_x clusters); Further addition of promoter would begin to cover up the iron surface such that reaction performance would decrease simply by there being fewer active sites.

The observed distinct discrepancy in the promotion effect of Mn is expected from a different phase transition of α -Fe₂O₃ to active phase in contrast to γ -Fe₂O₃. For the catalysts of Mn/ α -Fe₂O₃ and Mn/ γ -Fe₂O₃

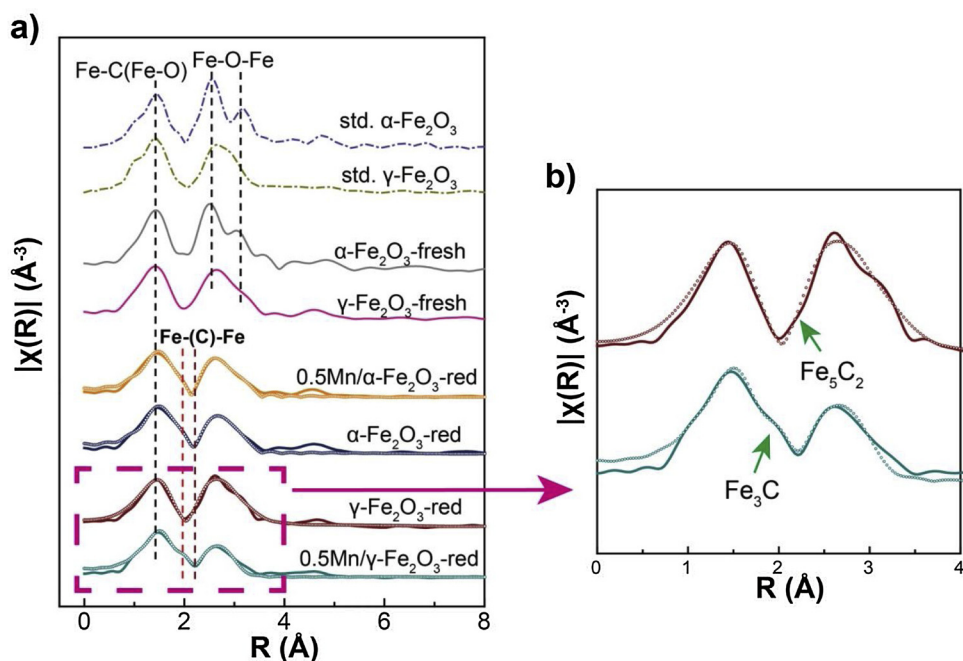


Fig. 6. The nonphase corrected, k^2 weighted Fourier-transformed *in situ* EXAFS spectra at the Fe K-edge of the $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ based catalysts after carburized. Dash lines denote reference samples of $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$, solid lines denote experiment spectra of iron based catalysts, and dotted lines denote fitting curves. The carburized samples were *in situ* treated in syngas ($\text{H}_2/\text{CO} = 1$, 15 ml/min) at 340 °C for 2 h at 1 bar.

after carburized/reaction, both $\chi\text{-Fe}_5\text{C}_2$ and $\theta\text{-Fe}_3\text{C}$ were formed as shown in XRD (Figs. S4–S6) and HRTEM results (Fig. S13). More importantly, the fraction of $\theta\text{-Fe}_3\text{C}$ phase on carburized/spent catalysts of 0.5 Mn/ $\gamma\text{-Fe}_2\text{O}_3$ is much higher than on 0.5 Mn/ $\alpha\text{-Fe}_2\text{O}_3$ (Tables 2 and 3). Compared to the absence of $\theta\text{-Fe}_3\text{C}$ on pure $\gamma\text{-Fe}_2\text{O}_3$ catalyst and its very low selectivity of olefins, addition of 0.5 wt. % Mn to $\gamma\text{-Fe}_2\text{O}_3$ largely increased selectivity of light olefins, which clearly demonstrated the promotion effect of Mn on $\gamma\text{-Fe}_2\text{O}_3$. It should be noted that this positive promotion effect on the formation of $\theta\text{-Fe}_3\text{C}$ phase was only observed obviously in $\gamma\text{-Fe}_2\text{O}_3$ based samples. In contrast, the addition of Mn on $\alpha\text{-Fe}_2\text{O}_3$ suppresses the carburization ability and leads to a lower total Fe_xC content (19.4% for carburized catalyst and 26.2% for spent catalyst) both in $\chi\text{-Fe}_5\text{C}_2$ and $\theta\text{-Fe}_3\text{C}$ phase (Tables 2 and 3). The results of Fe K-edge XANES (Fig. S14) underline this negative effect by Mn on $\alpha\text{-Fe}_2\text{O}_3$ based samples, as the limited Fe_xC formation was observed in Mn/ $\alpha\text{-Fe}_2\text{O}_3$. Similar results have also been reported in the literature which used $\alpha\text{-Fe}_2\text{O}_3$ or Fe_3O_4 [26,43]. Hence, we proposed

Table 4

Fe K-edge EXAFS fit results for the $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ nanorods materials after carburized.^a

Entry	Catalysts	Shell	CN	$R(\text{\AA})$	$\delta^2(\text{\AA}^{-2})$
1	$\gamma\text{-Fe}_2\text{O}_3$	Fe-O(C)	5.03 ± 0.93	1.98 ± 0.02	0.0172
		Fe-(C)-Fe	4.89 ± 1.81	2.77 ± 0.02	0.0149
		Fe-(O)-Fe	6.64 ± 2.26	3.25 ± 0.03	0.0255
2	0.5 Mn/ $\gamma\text{-Fe}_2\text{O}_3$	Fe-O(C)	3.10 ± 0.76	1.98 ± 0.03	0.0118
		Fe-(C)-Fe	1.32 ± 0.98	2.57 ± 0.05	0.0096
		Fe-(O)-Fe	4.22 ± 1.82	3.11 ± 0.03	0.0144
3	$\alpha\text{-Fe}_2\text{O}_3$	Fe-O(C)	3.54 ± 0.99	2.01 ± 0.04	0.0122
		Fe-(C)-Fe	0.44 ± 0.65	2.68 ± 0.02	0.0122
		Fe-(O)-Fe	4.48 ± 2.05	3.20 ± 0.01	0.0122
4	0.5 Mn/ $\alpha\text{-Fe}_2\text{O}_3$	Fe-O(C)	3.58 ± 0.99	2.01 ± 0.03	0.0034
		Fe-(C)-Fe	1.01 ± 0.66	2.71 ± 0.05	0.0034
		Fe-(O)-Fe	1.60 ± 1.00	3.21 ± 0.05	0.0034

^a The catalysts were pre-carburized at 340 °C, 0.1 MPa for 10 h using syngas ($\text{H}_2/\text{CO} = 1$).

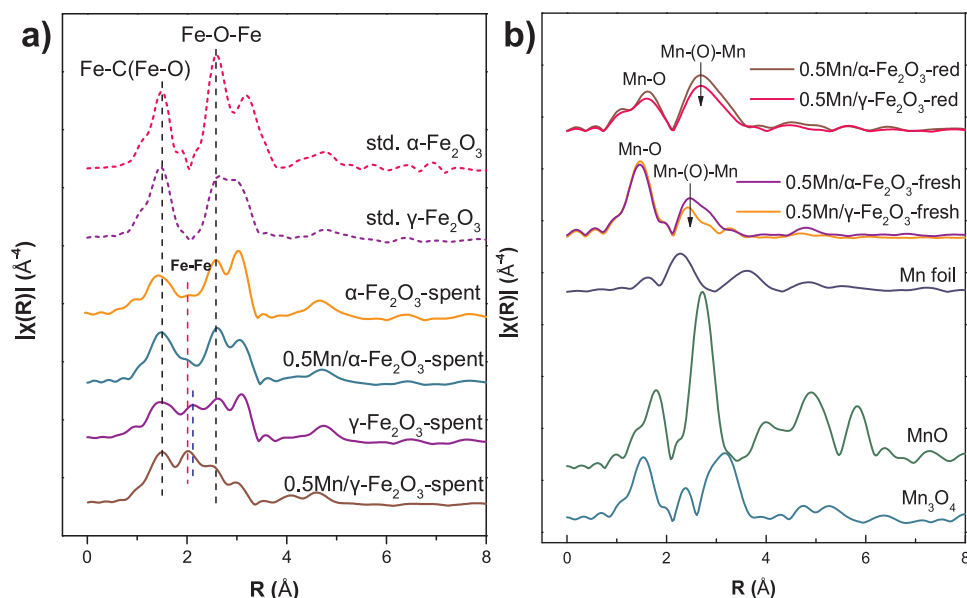


Fig. 7. a) The nonphase corrected, k^3 weighted Fourier-transformed *in situ* EXAFS spectra at the Mn K-edge of the $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ based catalysts after reaction. Dash lines denote reference samples of $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$, and solid lines denote experiment spectra of iron based catalysts. b) The nonphase corrected, k^3 weighted Fourier-transformed *in situ* EXAFS spectra at the Mn K-edge of the 0.5 Mn/ $\alpha\text{-Fe}_2\text{O}_3$ and 0.5 Mn/ $\gamma\text{-Fe}_2\text{O}_3$ nanorods catalysts after carburized. The carburized and spent samples were *in situ* treated in syngas ($\text{H}_2/\text{CO} = 1$, 15 ml/min) at 340 °C for 2 h at 1 bar.

Table 5
Catalytic activity and product selectivity of various iron-based catalysts at similar CO conversion.^a

Entry	Catalysts	W/F(CO + H ₂ + Ar) (g _{cat} h mol ⁻¹)	FTY (10 ⁻⁵ mol CO g _{Fe} ⁻¹ s ⁻¹)	CO ₂ (%CO conv.)	Hydrocarbon selectivity (c-mol %, CO ₂ -free)				O/P	Exit H ₂ /CO molar ratio
					CH ₄	C ₂₋₄	C ₂₋₄	C ₅₊		
1	γ-Fe ₂ O ₃	4.34	2.98	38.2	21.4	24.8	25.2	28.6	0.98	1.47
2	0.5 Mn/γ-Fe ₂ O ₃	5.83	2.31	36.8	10.9	58.4	9.5	21.2	6.15	1.28
3	α-Fe ₂ O ₃	6.02	2.22	35.9	14.9	41.6	10.7	32.8	3.89	1.33
4	0.5 Mn/α-Fe ₂ O ₃	10.9	1.25	37.1	13.7	36.7	14.6	35	2.51	1.45

^a Reaction condition: 0.2 g catalyst, 320 °C, 1.0 MPa, H₂/CO = 1, time on stream 20 h. Data were reported at CO conversion levels between 63% and 68%, and were taken at 20 h of time on stream.

that the slightly higher light olefins selectivity of 0.5 Mn/α-Fe₂O₃ catalyst was caused by the much lower hydrogenation capacity (CO Conv. 29.4%) than nonpromoted α-Fe₂O₃ catalyst (CO Conv. 52.7%). To confirm this speculation, additional tests were carried out at 1 MPa, 320 °C, and an H₂/CO ratio of 1 to compare the products selectivity of catalysts at similar CO conversion levels (63–68 %, Table 5). As expected, the 0.5 Mn/α-Fe₂O₃ catalyst only achieved a light olefins selectivity of 36.7%, which is lower than nonpromoted α-Fe₂O₃ catalyst (41.6%). Meanwhile, the CO₂ selectivity for all the samples was approximately 35–39 % on the basis of CO converted. Since the CO conversion and CO₂ selectivity of four catalysts is at a comparative level, this leads to the little difference in exit molar H₂/CO ratio at a range from 1.28 to 1.47.

When Mn modified, Fig. 6b shows that the first Fe-(C)-Fe shell distance of 0.5 Mn/γ-Fe₂O₃ after carburized is much shorter (1.98 Å, uncorrected) than γ-Fe₂O₃, due to the contributions from θ-Fe₃C (21.0% in Mössbauer spectroscopy). The CNs of Fe-C(O) and Fe-Fe in the iron carbide are 3.10 and 1.32, respectively (Table 4). In contrast, for the carburized 0.5 Mn/α-Fe₂O₃ catalyst, the first Fe-(C)-Fe shell distance in iron carbides is similar to the unpromoted α-Fe₂O₃ catalyst, verifying that Mn had no promotion effect toward formation of θ-Fe₃C from α-Fe₂O₃. More importantly, the fitting results (Table 4) show that the Fe-Fe bond length in 0.5 Mn/γ-Fe₂O₃ is even shorter (2.57 Å) than in 0.5 Mn/α-Fe₂O₃ (2.71 Å), due to the higher proportion of θ-Fe₃C. Since *k*³-weighted *r*-space spectrum is more sensitive in the high *Z* element signal contribution at high *k* value [44], here the *k*³-weighted spectra are also summarized in Fig. S15 for a better comparison. And the differences between *r*-space spectra in *k*²-weighted spectra remain in the *k*³-weighted spectra.

Based on the above results, we proposed that the formation of surface iron species was influenced by Mn at different degrees on α-Fe₂O₃ and γ-Fe₂O₃ nanorods, which leads to the different product selectivity especially for light olefins. To probe the surface electronic character evolution of Fe catalysts during FTS reaction, the *in situ* synchrotron radiation XPS features in Fe 2p and Mn 2p region were monitored for all catalysts (Fig. 8). The catalysts were treated in syngas (H₂/CO = 1) at 340 °C for 2 h at 1 bar, after treatment the samples were transferred to the measurement chamber without exposure to air. For the fresh sample, all catalysts show a characteristic Fe³⁺ and Mn³⁺ 2p_{3/2} peak at ~711.0 eV and ~641.6 eV, respectively [45,46]. Meanwhile, Mn promoted γ-Fe₂O₃ samples (Fig. 8a) possess a lower binding energy of Fe than the corresponding nonpromoted samples (γ-Fe₂O₃) and Mn/α-Fe₂O₃ (Fig. 8a), demonstrating the presence of significant Fe-Mn interaction in this unique catalyst structure [47,48], i.e., MnO_x supported on nonporous γ-Fe₂O₃ nanorods without any other inert supports. Moreover, it has been reported that γ-Fe₂O₃ has an inverse spinel structure with cation vacancies, which may adsorb metal to form some special/stronger interaction [18]. Hence, we proposed that MnO_x species locate on the surface of γ-Fe₂O₃ nanorods and form strong interaction with clear metal-promoter interfaces and close proximity to Fe. The extended X-ray absorption fine structure (EXAFS) studies clarified that the MnO_x on γ-Fe₂O₃ nanorods possessed less Mn-Mn

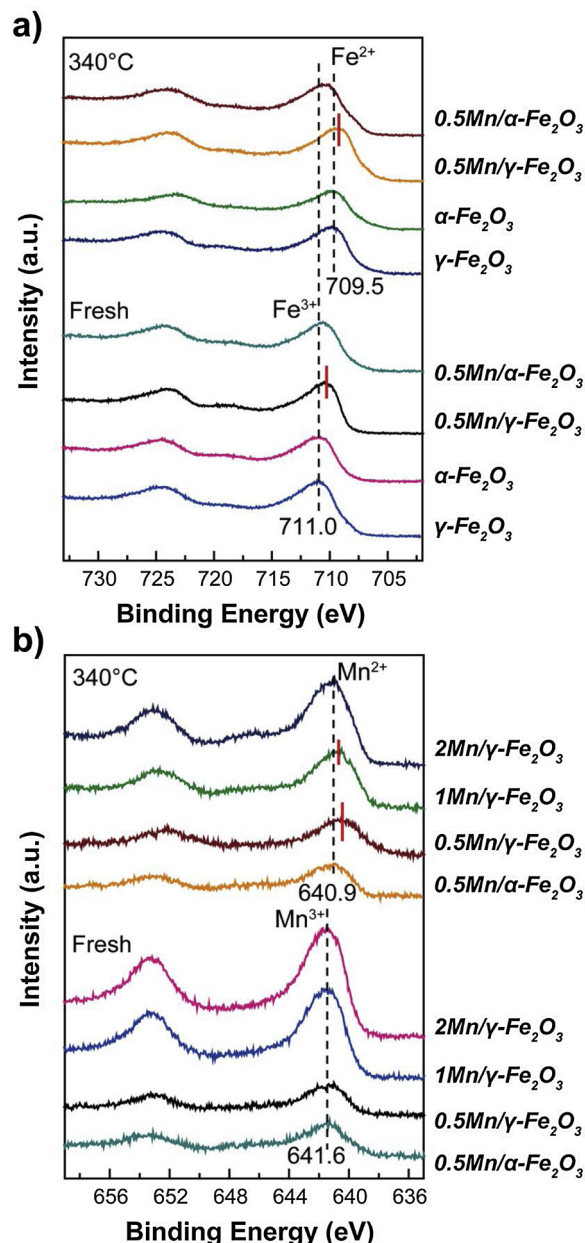


Fig. 8. The *in situ* synchrotron radiation XPS spectra of Fe 2p (a) and Mn 2p (b) for fresh and spent catalysts. All spent samples were *in situ* treated in syngas (H₂/CO = 1, 15 ml/min) at 340 °C for 2 h at 1 bar.

coordination than on the corresponding α-Fe₂O₃ nanorods as prepared and after carburized (Fig. 7b), suggesting the Mn/γ-Fe₂O₃ nanorods catalyst had more Mn edge sites [49] due to the strong Fe-Mn

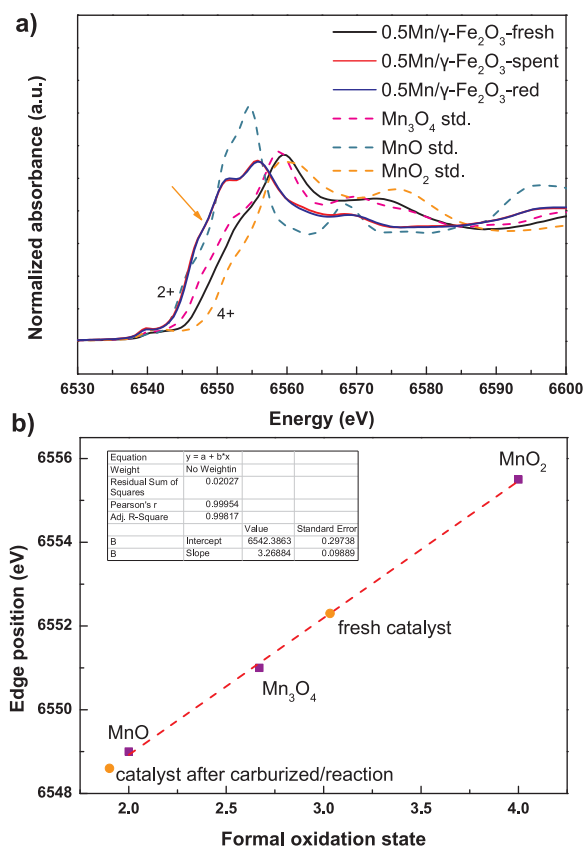


Fig. 9. a) *in situ* X-ray absorption near edge spectra (XANES) at Mn K-edge of 0.5 Mn/γ-Fe₂O₃ nanorods catalysts as-prepared, after carburized, and after reaction, in comparison with references for MnO₂, MnO, and Mn₃O₄. All the pre-carburized catalysts were carburized again *in situ* in syngas (CO/H₂ = 1) at 340 °C for 1 h; b) Correlation between XANES absorption edge position of Mn K-edge and oxidation state of Mn in 0.5 Mn/γ-Fe₂O₃ samples as prepared, after carburized and after reaction.

interaction.

After reduction, the peaks in Fe 2p and Mn 2p regions both shifted toward lower binding energies that correspond to the formation of Fe²⁺ and Mn²⁺ species [50,51]. Furthermore, the Mn 2p peaks of Mn/γ-Fe₂O₃ possess a more reduced state of Mn compared with Mn/α-Fe₂O₃ sample (Fig. 8b). Among them 0.5 Mn/γ-Fe₂O₃ exhibited lowest binding energy, which is more than 0.5 eV lower (640.4 vs. 640.9 eV). This result suggests that more reduced MnO_{1-x} species was formed on γ-Fe₂O₃ supported samples. It is proposed that the interfacial MnO_x with more low-coordinated edge sites might be more easily reduced by syngas, due to the higher intrinsic reactivity on edge sites [52–54] and interfacial spillover effect [8].

The normalized Mn K-edge X-ray absorption near-edge structure (XANES) for 0.5 Mn/γ-Fe₂O₃ and 0.5 Mn/α-Fe₂O₃ samples as prepared, after carburized and after reaction was further performed Figs. 9 and 10). It can be seen that the Mn oxidation state of fresh samples was both Mn³⁺, which is evidenced by the absorption edge position of Mn K-edge (Fig. 9b). The shift of the absorption edge toward energies closer to Mn²⁺ (MnO reference) edge (6549 eV) after reduction show that Mn is further reduced during carbonization process, and the same absorption edge after reaction suggests that it maintained stable valence under catalytic conditions. Furthermore, Mn absorption edge in 0.5 Mn/γ-Fe₂O₃ after carburized at 340 °C is at lower energy than that of pure MnO reference, which further confirmed that Mn species is presents as a more reduced chemical state than a pure MnO [8].

Iron carbides are classified as interstitial carbides [55], in which carbon atoms are placed in available interstitial sites of Fe⁰ atoms. The

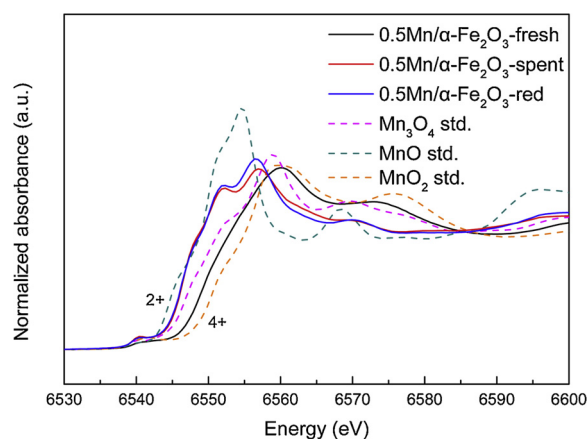


Fig. 10. *In situ* XANES at Mn K-edge of 0.5 Mn/α-Fe₂O₃ nanorods catalysts as-prepared, after carburized, and after reaction, in comparison with references for MnO₂, MnO, and Mn₃O₄. All the pre-carburized catalysts were carburized again *in situ* in syngas (CO/H₂ = 1) at 340 °C for 1 h.

present studies indicate that the presence of strong Fe-Mn interaction suppress diffusion of C atoms into the lattice of Fe oxide such that C-poor phases (cementite) emerge for Mn/γ-Fe₂O₃ catalysts. The *r*-space structural parameters (Table 4) show that the Fe-(O)-Fe bond length shrinks in 0.5 Mn/γ-Fe₂O₃ (3.11 Å) compared to that of γ-Fe₂O₃ (3.25 Å). We believe that it originates from the lattice distortion in the surface Fe layers [56] at interfacial boundary of Fe_xO_y-MnO_x, which strong interact with MnO_{1-x}. Hence, the incorporation of carbon to iron lattices will be suppressed, due to the kinetic factors limitation [10,14]. As a result, the formation of θ-Fe₃C phase which have weak hydrogenation activity for unsaturated hydrocarbon was promoted. Meanwhile, the special morphology structure of catalyst (promoter locate on nonporous iron oxide nanorods) would reduce the diffusion resistance of products, especially for light olefins. Hence, the selectivity of light olefins on Mn/γ-Fe₂O₃ catalysts was improved dramatically. For Mn/α-Fe₂O₃ catalysts, selectivity of light olefins increased only slightly. This little improvement can be ascribed to the lower hydrogenation capacity (*i.e.* FTS activity) on Mn-modulated α-Fe₂O₃ catalyst, as the total Fe_xC content was decreased (from 25.1% to 19.4%, Table 2).

4. Conclusion

Correlation among the original phase of iron oxide and the corresponding catalytic performance is crucial for designing a catalyst with high selectivity in STO reaction. Pure α-Fe₂O₃ and γ-Fe₂O₃ form quite different iron carbides and thus exhibit different catalytic selectivity for light olefins. By loading Mn on the surface of Fe₂O₃, light olefins selectivity was largely promoted on γ-Fe₂O₃ but not on α-Fe₂O₃. This difference in promoting catalytic selectivity is closely related to structure of original iron oxide crystal structure and formation of iron carbides. Fundamental understanding of the intrinsic correlation between original oxide structure and the formation of iron carbide and promotion effect of Mn was achieved through *in situ* and *ex situ* characterizations.

In summary, with the goal of understanding the effects of initial iron oxide crystal phase on the active phase and controlling FTS performance, we prepared α-Fe₂O₃ and γ-Fe₂O₃ nonporous nanorods to explore the iron phase evolution during FTS, which can avoid the influence of diffusion limitation on the selectivity during the FTS reaction. Our studies reveal that the formation of high selective iron carbides (θ-Fe₃C) to light olefins was affected by initial iron oxide crystal phase. The more reduced MnO_{1-x} edge sites with strong Fe-Mn interaction was preferentially formed on the surface of γ-Fe₂O₃ nanorods, which was beneficial to the formation of C-poor iron carbides phase (θ-Fe₃C). Consequently, due to the morphology effect (promoter locate on

nonporous nanorods) and crystal phase effect (γ -Fe₂O₃), Mn-modulated γ -Fe₂O₃ nanorods catalysts can achieve light olefins selectivity of up to 50.1–61.2 % with more than 50% of CO single-pass conversion under industrially relevant conditions. This catalyst exhibits high stability for at least 50 h on stream.

Declaration of Competing Interest

The authors declare no competing financial interest.

Acknowledgments

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.apcatb.2019.118219>.

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Update

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Corrigendum

Corrigendum to “Effects of initial crystal structure of Fe_2O_3 and Mn promoter on effective active phase for syngas to light olefin” [Appl. Catal. B: Environ. 261 (2020) 118219]



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The authors regret to inform that the following error in the aforementioned article. The HRTEM image in Fig. 4b was incorrect. Fig. 4b should be corrected to

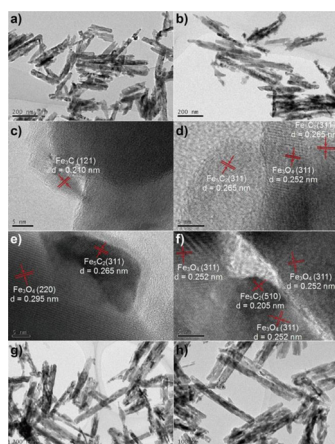


Fig. 4. HRTEM of $\alpha\text{-Fe}_2\text{O}_3$ (a, c, e, g) and $\gamma\text{-Fe}_2\text{O}_3$ (b, d, f, h) nanorods after carburized (a–f) and after FTS reaction (g, h). Carburized conditions: 340 °C, 0.1 MPa, $\text{H}_2/\text{CO} = 1$, $\text{W/F}(\text{CO} + \text{H}_2 + \text{Ar}) = 2.5 \text{ g}_{\text{cat}} \cdot \text{h} \cdot \text{mol}^{-1}$. Reaction conditions: 320 °C, 1.0 MPa, $\text{H}_2/\text{CO} = 1$, $\text{W/F}(\text{CO} + \text{H}_2 + \text{Ar}) = 5 \text{ g}_{\text{cat}} \cdot \text{h} \cdot \text{mol}^{-1}$.

Above correction do not change the overall results & discussion, and conclusion reported in the published article.

The authors would like to apologise for any inconvenience caused.

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Supporting Information

Effects of Initial Crystal Structure of Fe₂O₃ and Mn Promoter on Effective Active Phase for Syngas to Light Olefins

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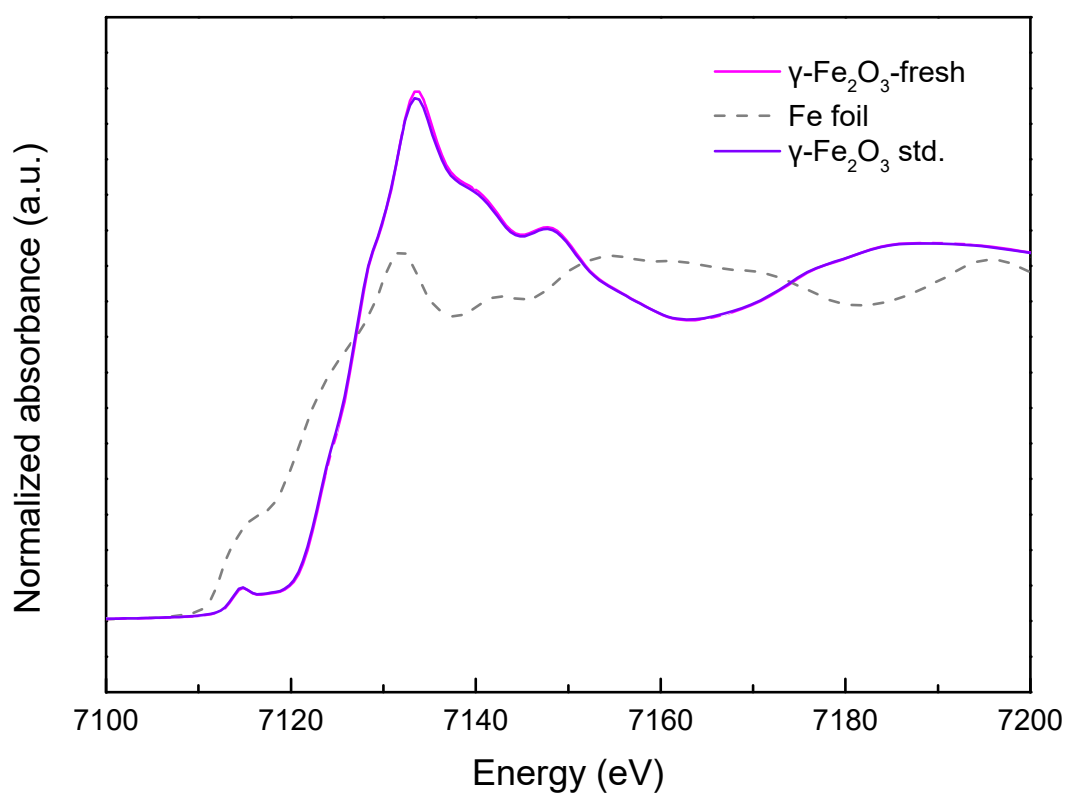


Figure S1. Fe K-edge XANES of fresh $\gamma\text{-Fe}_2\text{O}_3$ catalyst and Fe standards.

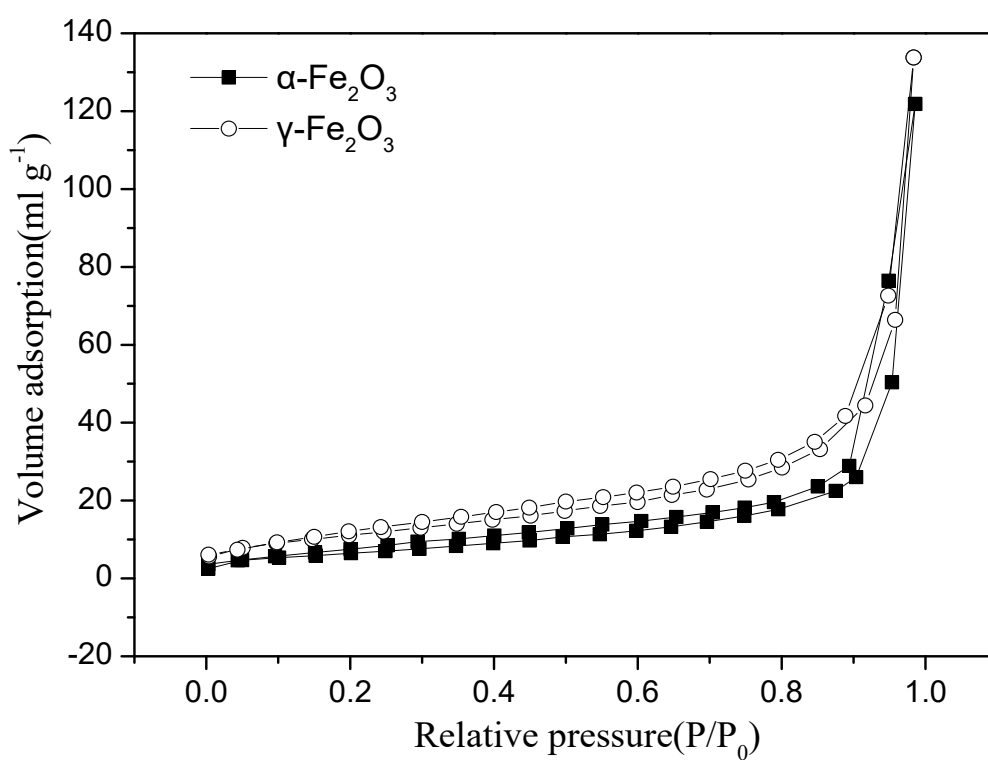


Figure S2. N_2 adsorption-desorption isotherms of $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ nanorods as prepared.

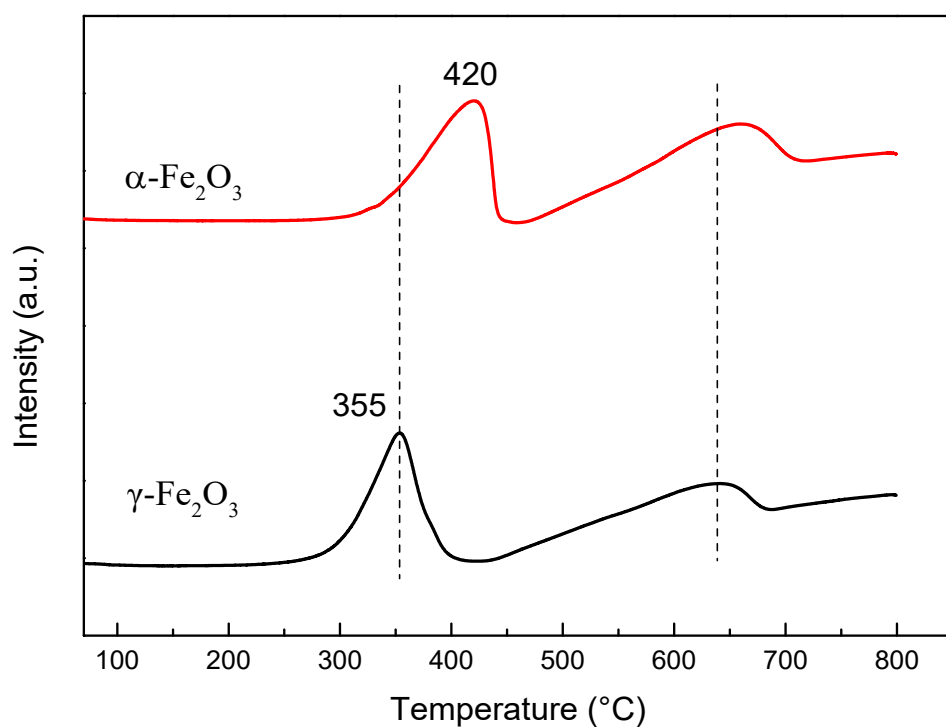


Figure S3. H₂-TPR profiles of $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ nanorods catalysts.

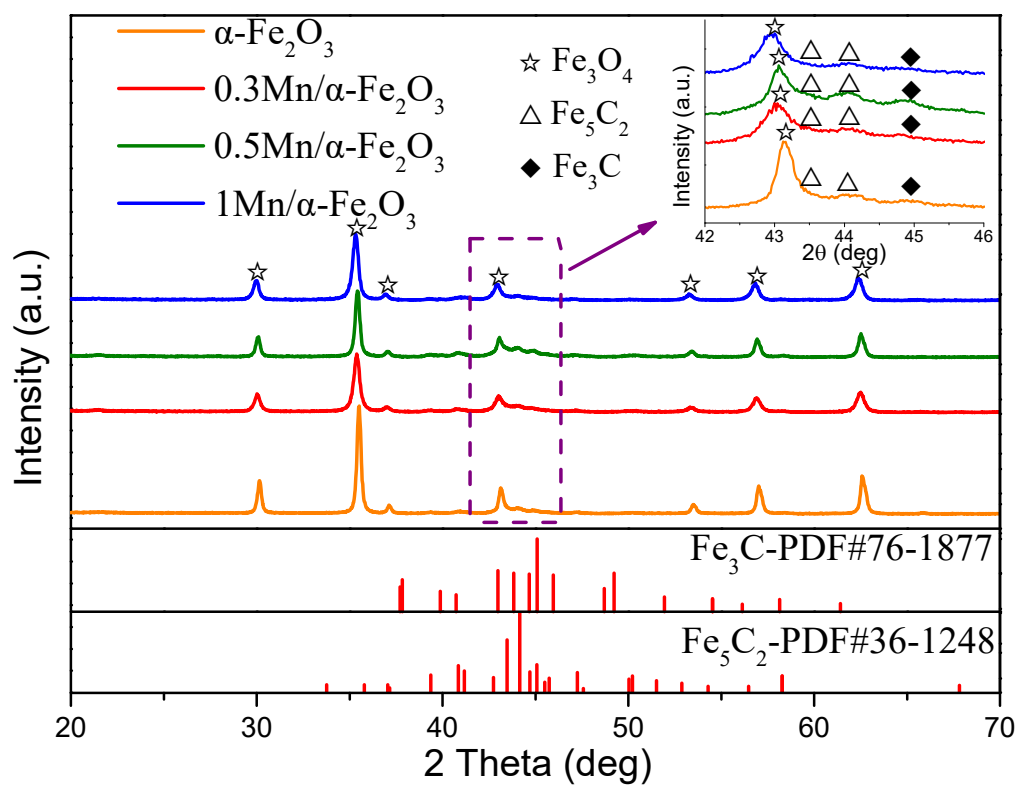


Figure S4. XRD patterns for the $\alpha\text{-Fe}_2\text{O}_3$ nanorods and Mn promoted samples after carburization.

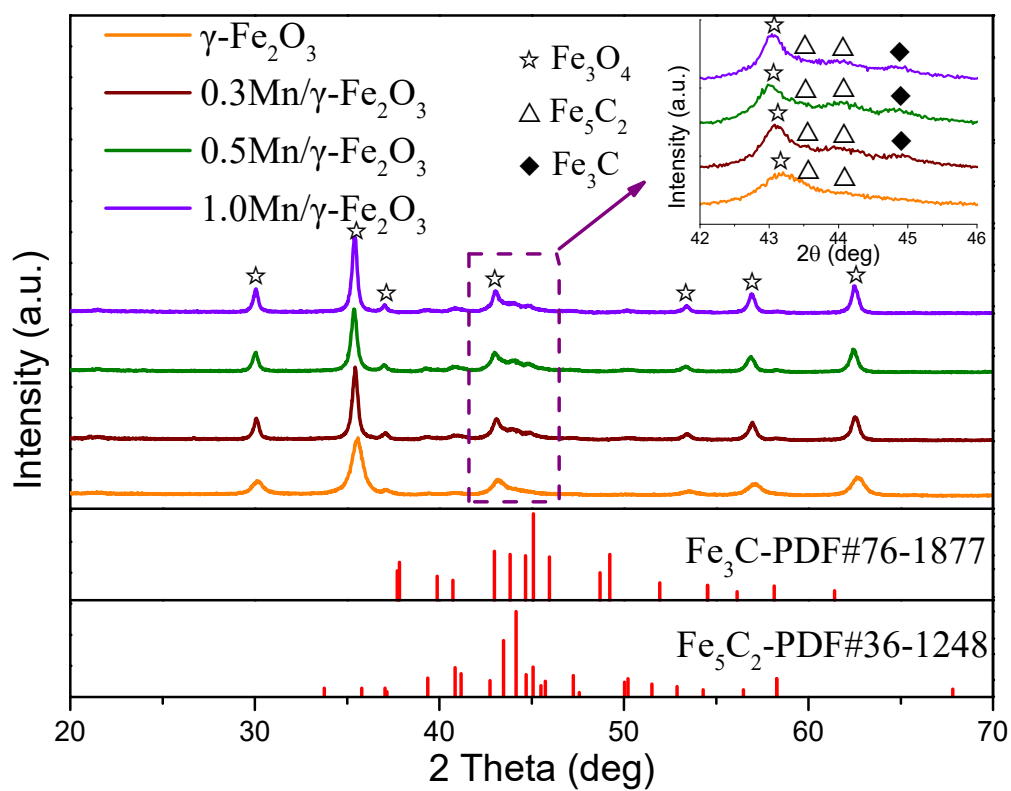


Figure S5. XRD patterns for the $\gamma\text{-Fe}_2\text{O}_3$ nanorods and Mn promoted samples after carburization.

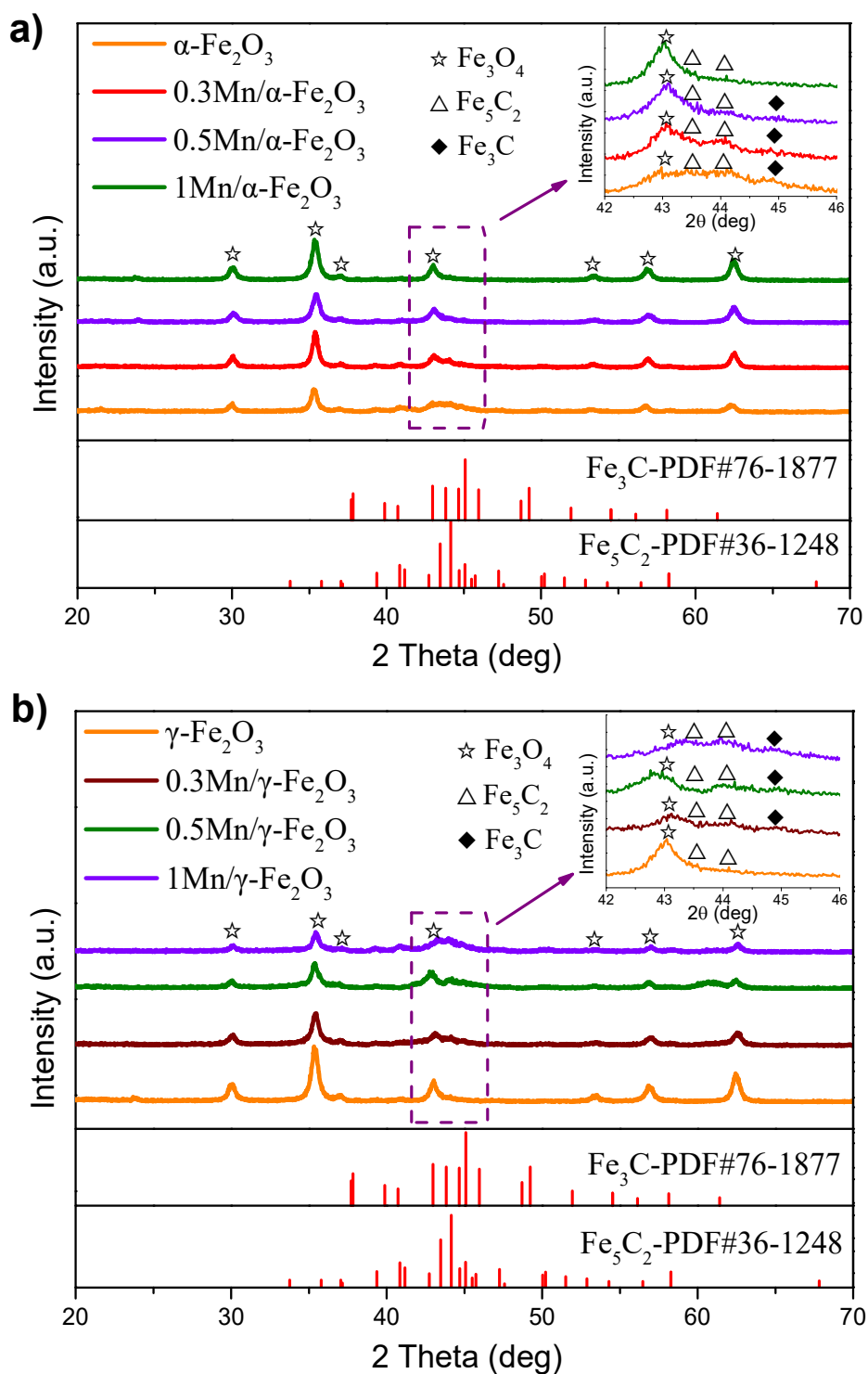


Figure S6. a) XRD patterns of $\alpha\text{-Fe}_2\text{O}_3$ nanorods and Mn promoted samples after 20 h reaction. b) XRD patterns of $\gamma\text{-Fe}_2\text{O}_3$ nanorods and Mn promoted samples after 20 h reaction.

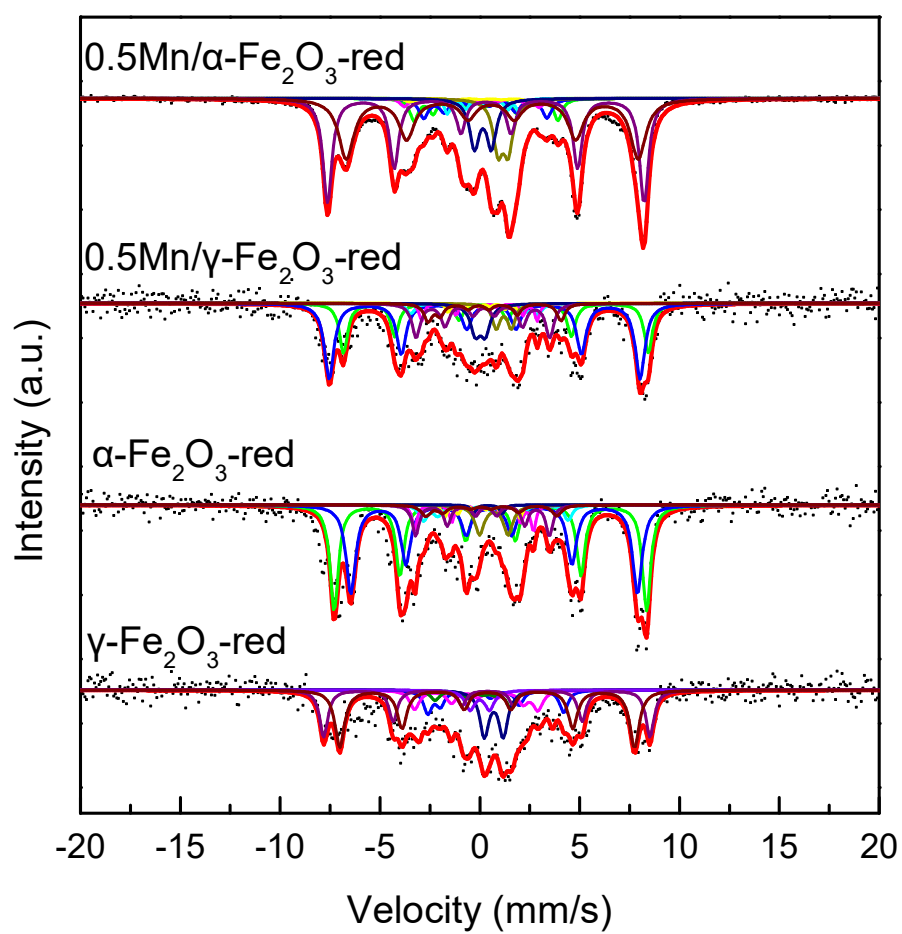


Figure S7. Mössbauer spectra of α -Fe₂O₃ and γ -Fe₂O₃ nanorods catalysts after carburization.

Carburization conditions: 340 °C, 0.1MPa, H₂/CO=1, W/F(CO+H₂+Ar) = 2.5 g_{cat}·h·mol⁻¹.

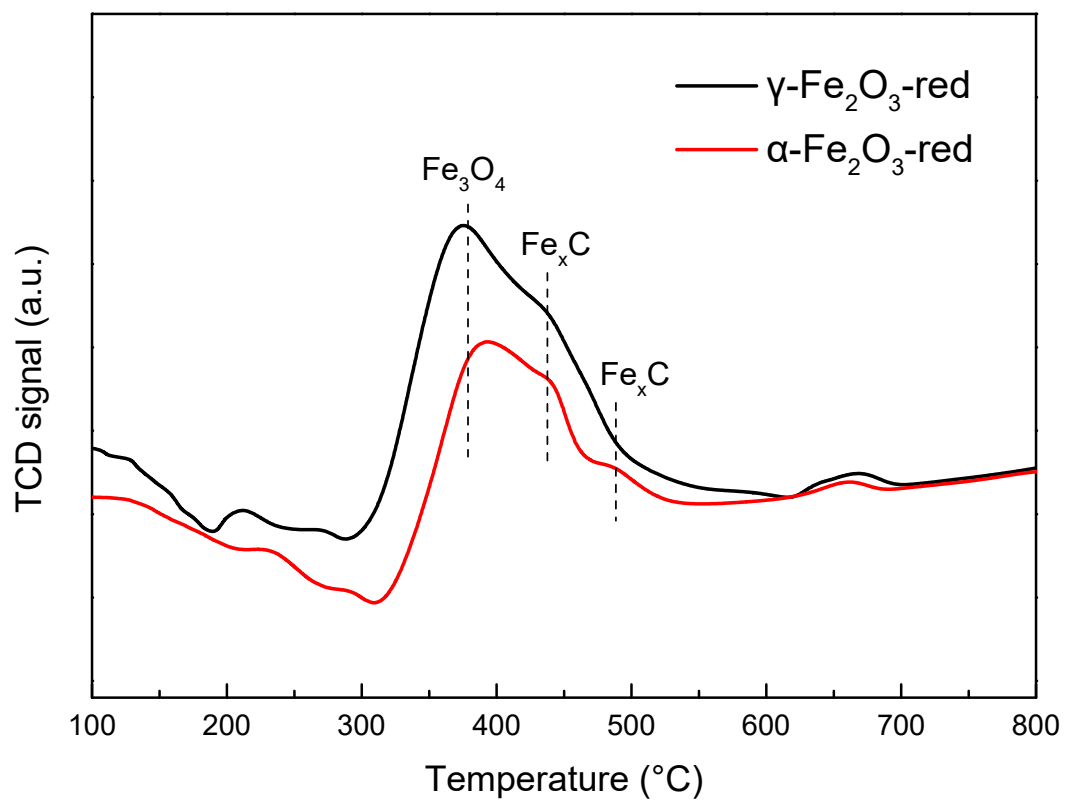


Figure S8. CO-TPD profiles of $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ nanorods catalysts after carburization.

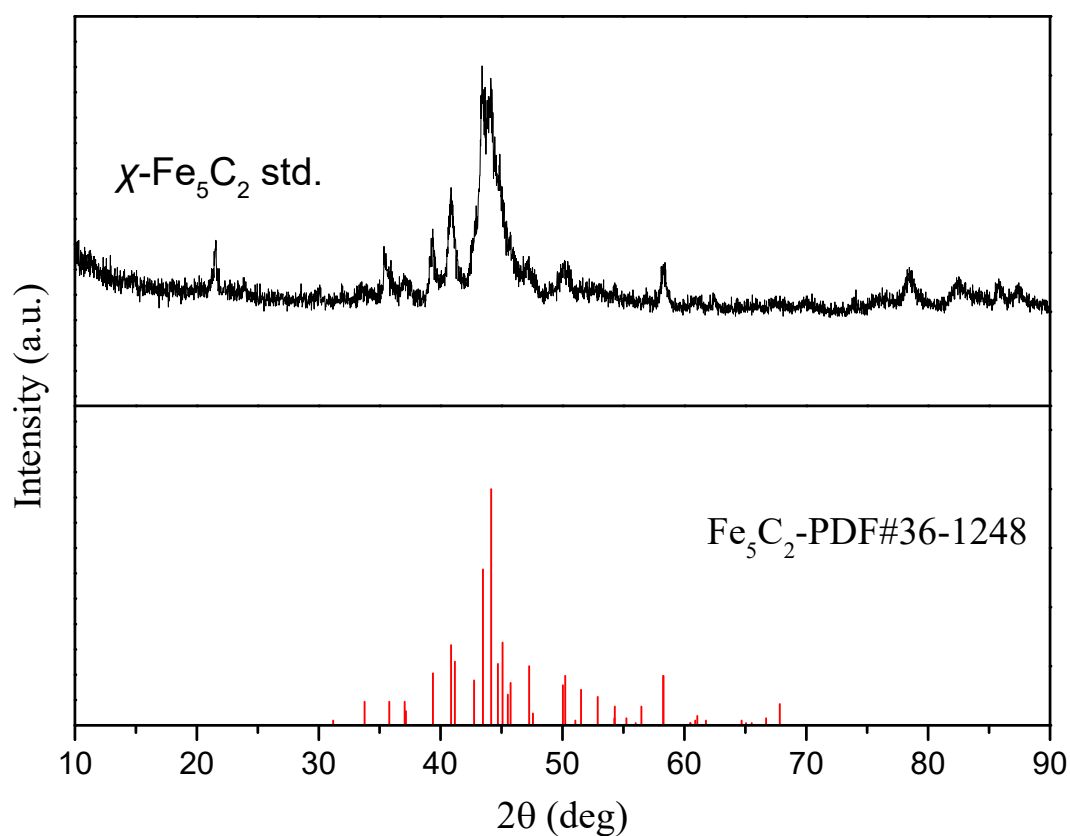


Figure S9. XRD spectra of χ - Fe_5C_2 reference sample as prepared by in situ carbonization of hematite.

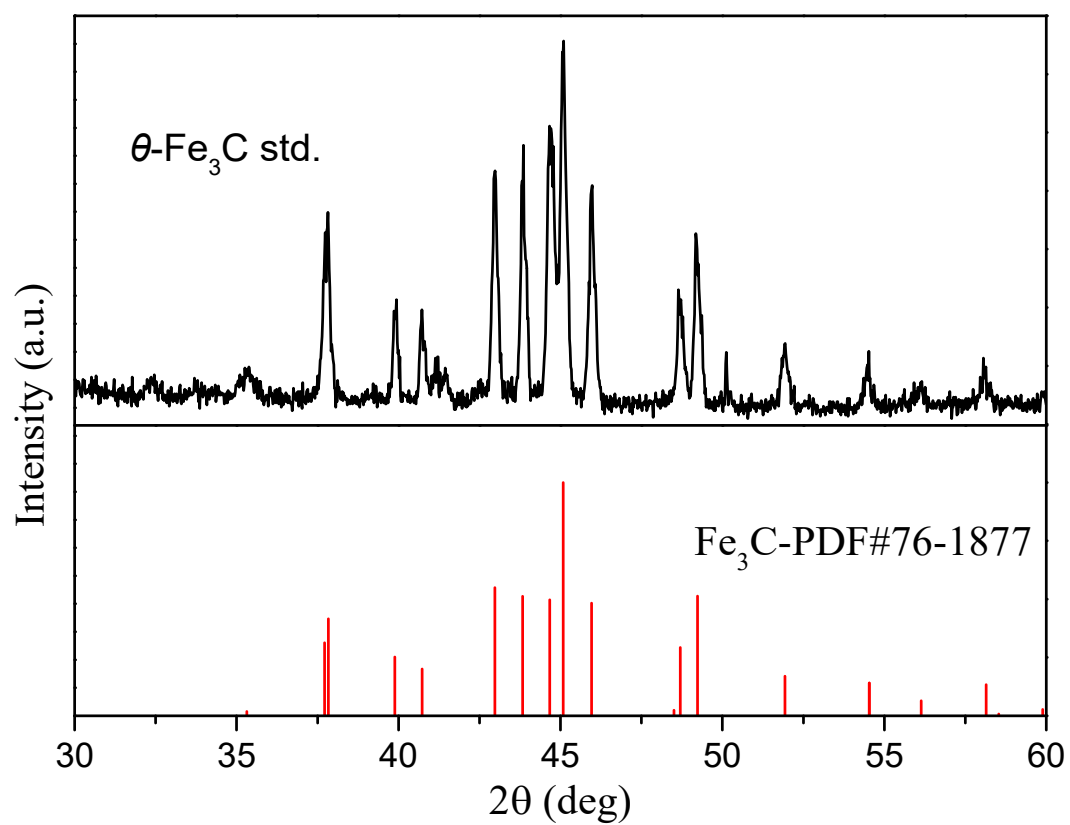


Figure S10. XRD spectra of θ - Fe_3C reference sample as prepared by wet-chemical route.

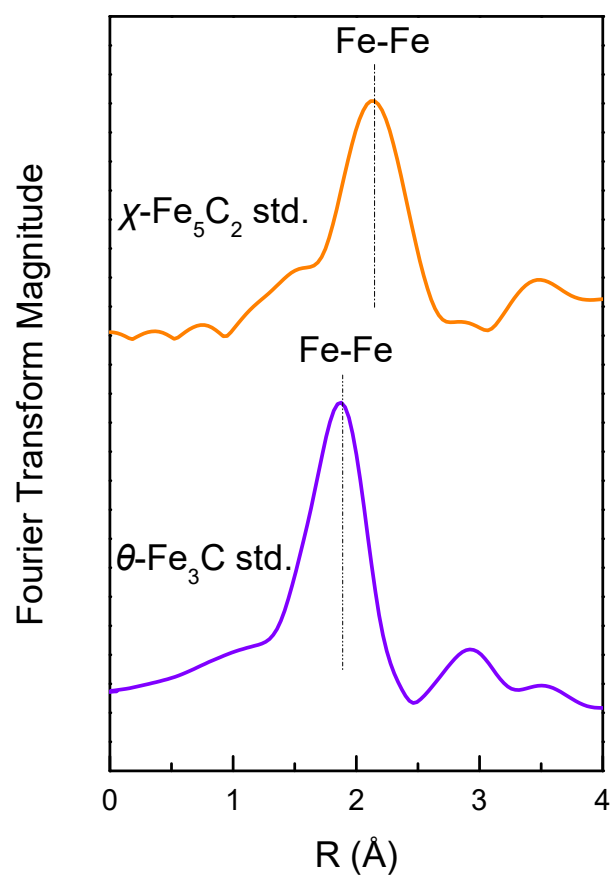


Figure S11. The nonphase corrected, Fourier transformed (FT) Fe K-edge EXAFS spectra of χ -Fe₅C₂ and θ -Fe₃C reference samples.

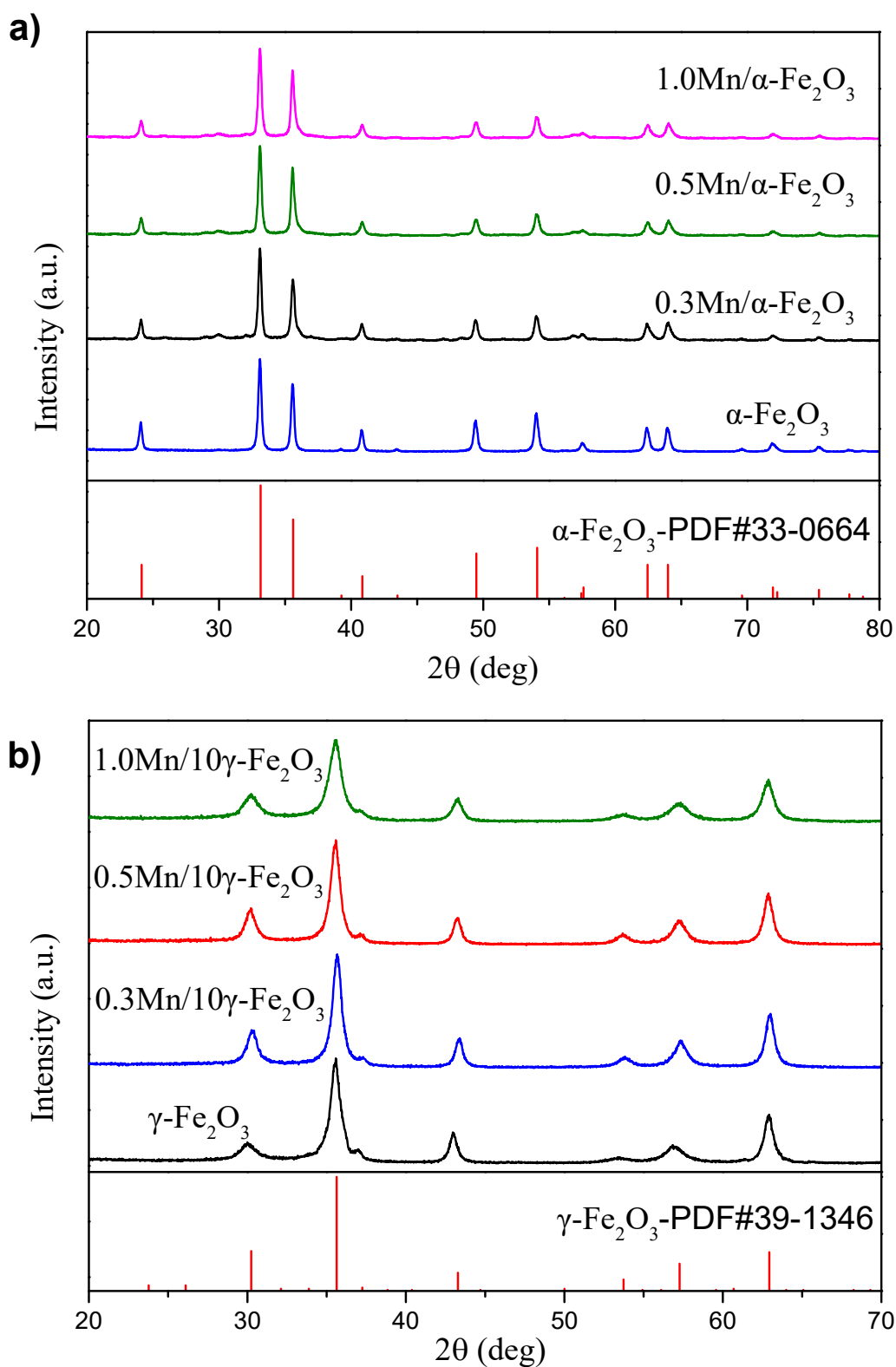


Figure S12. X-ray diffraction of α -Fe₂O₃ (a) and γ -Fe₂O₃ (b) nanorods catalysts with different manganese loading as prepared.

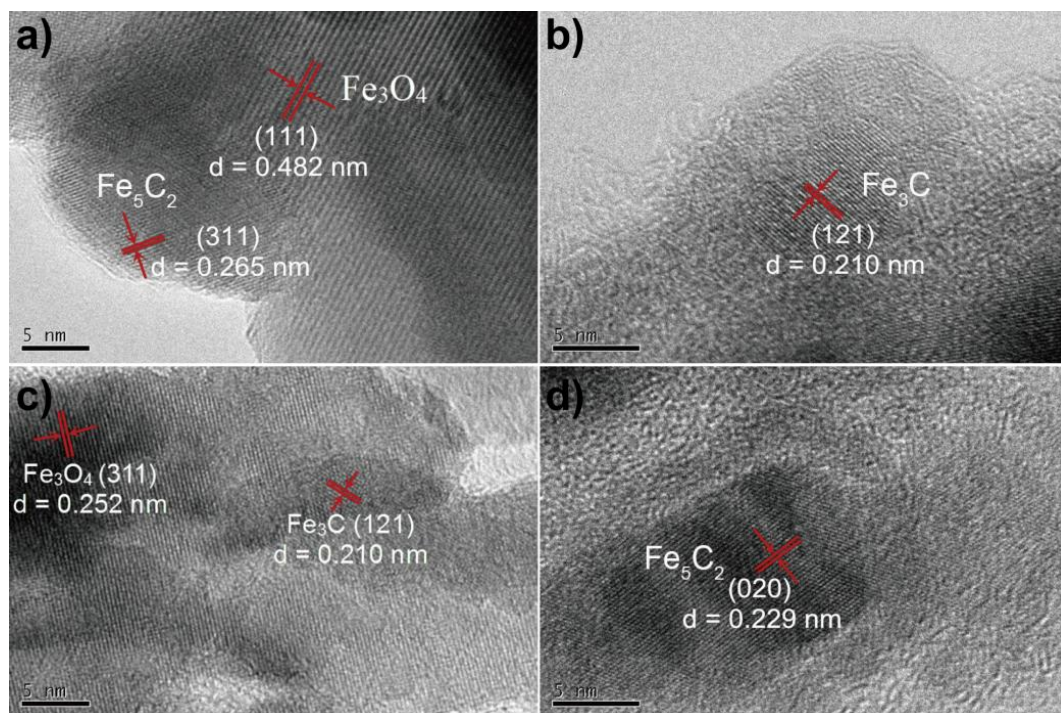


Figure S13. HRTEM of 0.5Mn/ α -Fe₂O₃ (a, b) and 0.5Mn/ γ -Fe₂O₃ (c, d) nanorods catalysts after carburization. Carburization conditions: 340 °C, 0.1 MPa, H₂/CO=1, W/F(CO+H₂+Ar) =2.5 g_{cat}·h·mol⁻¹.

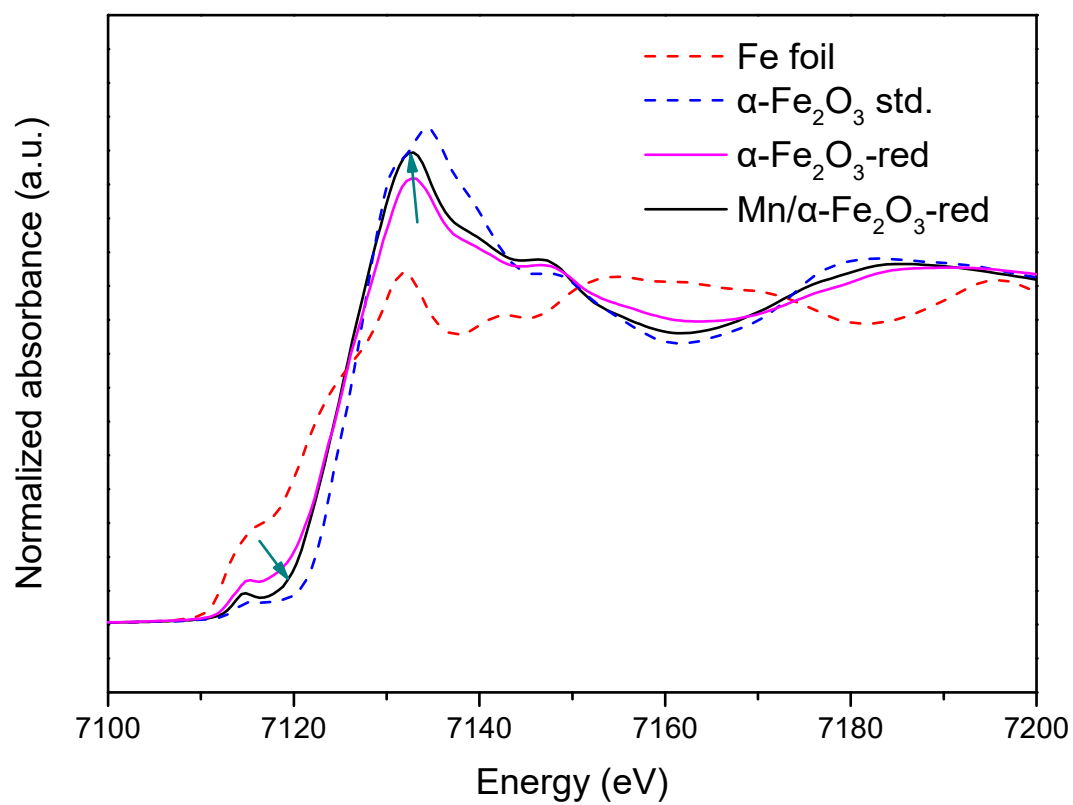


Figure S14. *In situ* Fe K-edge XANES of 0.5Mn/ α -Fe₂O₃ catalyst and Fe standards. The pre-carburized catalysts were carburized again in-situ in syngas (CO/H₂=1) at 340 °C for 1 h.

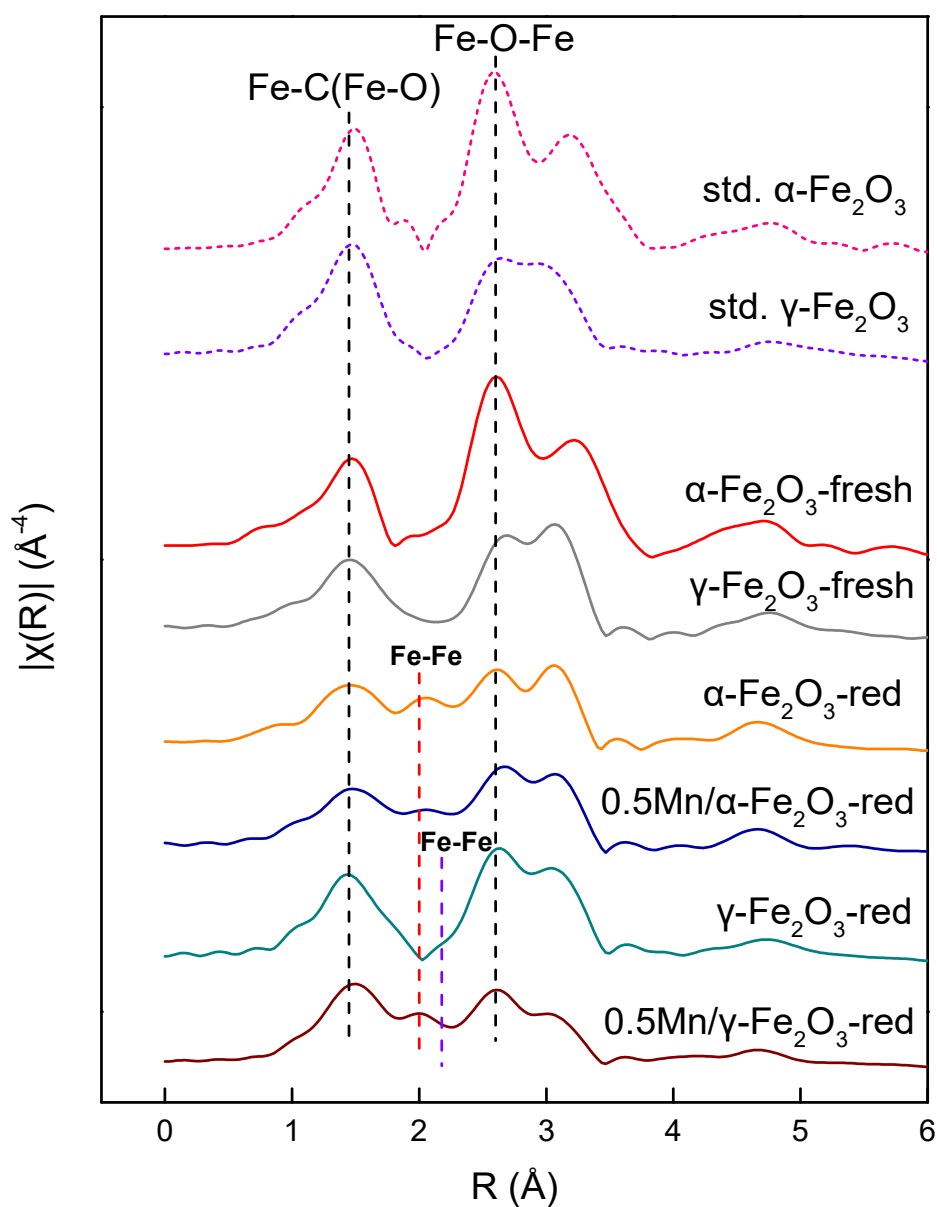


Figure S15. The nonphase corrected, k^3 weighted Fourier-transformed EXAFS spectra at the Fe K-edge of the $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$ based catalysts as-prepared and after carburized. Dash lines denote reference samples of $\alpha\text{-Fe}_2\text{O}_3$ and $\gamma\text{-Fe}_2\text{O}_3$, and solid lines denote experiment spectra of iron based catalysts. The carburized samples were in situ treated in syngas ($\text{H}_2/\text{CO}=1$, 15 ml/min) at 340 °C for 2 h at 1 bar.

Table S1 Least squares fitting of XANES data for various spent iron-based catalysts.

Catalysts	Fe ₂ O ₃	Fe ₃ O ₄	Fe _x C ^a
γ-Fe ₂ O ₃	-	63.3 %	36.7 %
0.5Mn/γ-Fe ₂ O ₃	-	56.1 %	43.9 %
α-Fe ₂ O ₃	2.8 %	75.8 %	31.4 %
0.5Mn/α-Fe ₂ O ₃	7.9 %	70.0 %	22.1 %

^a Total content of Fe₅C₂+Fe₃C